

Welcome to [The Ultimate 11+ Maths Strategy Guide](#)!

I am super excited to share this book which, after months, has come to fruition. It ultimately is designed to provide you with a step-by-step approach to preparing for the 11+ Maths exam, whether you're a parent working with your child at home, or a tutor helping students refine their skills. The strategies shared here have been honed through years of tutoring experience, working with hundreds of students, and carefully analysing dozens of past GL Maths papers. What you'll find inside are not just answers to questions, but unique, tailored strategies to tackle every possible question type you might encounter in the exam.

How to Use This Book:

- 1. Start with the Basics:** If you're unfamiliar with the 11+ or want to refresh your knowledge, begin with the introduction and general principles. This section will give you an overview of the exam structure, types of questions, and how best to approach preparation.
- 2. Dive into the Strategy Sections:** The book is divided into clear sections, each focusing on a specific question type or topic area (e.g., Fractions, Percentages, Word Problems, etc.). You can either read through the sections in order or jump to the particular question types that you or your child find more challenging.
- 3. Follow the Step-by-Step Strategies:** Each strategy is written to be intuitive and easy to follow. I've broken down even the most complex question types into clear, actionable steps. These strategies are unique to each question type, so you'll never be left wondering how to approach something new. Take your time, and make sure to work through the examples thoroughly.
- 4. Practice with Purpose:** After each strategy, you'll find a set of practice questions. These are designed to help you build confidence and reinforce the skills you've just learned. Practice is key, and these questions are specifically chosen to reflect the types of problems your child will face in the exam.
- 5. Review the Solutions:** For each set of practice questions, I've included detailed solutions. These aren't just answers—they're thorough explanations of the reasoning behind each step. When reviewing the solutions, pay special attention to the methods and strategies used to arrive at the correct answer.
- 6. Track Progress:** As you work through the book, make note of any areas that feel particularly tricky. Once you've mastered the strategies, go back and reattempt questions from those sections. Consistent practice is essential, and over time, you'll notice significant improvements.

1. Mental Arithmetic & Quick Calculation Strategies

In the GL Maths exam, mental arithmetic is a vital section, as it assesses how well students can perform basic arithmetic operations under time pressure. When I first start working with students, I focus on speed, accuracy, and confidence in mental calculations.

Here are strategies I use to ensure my students ace these fundamentals.

Strategy 1: Break Down Large Numbers into Simpler Parts (Decomposition)

One of the key techniques I teach is the decomposition method, which involves breaking larger numbers down into smaller, more manageable components. This helps with speed and accuracy.

Why this Works:

Breaking a large number into hundreds, tens, and ones allows students to perform simpler calculations step-by-step. It takes advantage of the distributive property of numbers, making complex problems easier to solve.

Example 1: $762 + 348$

Let's break this down in detail.

1. Decompose the numbers:

- $762 = 700 + 60 + 2$
- $348 = 300 + 40 + 8$

2. Add the corresponding parts together:

- Add the hundreds: $700 + 300 = 1000$
- Add the tens: $60 + 40 = 100$
- Add the ones: $2 + 8 = 10$

3. Combine the results:

- $1000 + 100 + 10 = 1110$

Encourage your child to always break down numbers when faced with addition problems. With practice, they will become faster at it.

Example 2: $1,234 + 689$

1. Decompose the numbers:

- $1,234 = 1,200 + 30 + 4$
- $689 = 600 + 80 + 9$
- 2. Add the parts:
 - $1,200 + 600 = 1,800$
 - $30 + 80 = 110$
 - $4 + 9 = 13$
- 3. Add all parts:
 - $1,800 + 110 + 13 = 1,923$

Strategy 2: Use Rounding to Simplify Calculations

This strategy helps when adding or subtracting numbers that are close to multiples of 10, 100, or 1,000. Instead of working with awkward numbers, we round them to the nearest multiple and adjust after.

Why this Works:

Rounding numbers makes mental calculations faster, but you need to adjust the result afterwards. It works best for addition and subtraction.

Example 1: $517 + 432$

1. Round to nearest tens:
 - Round 517 to 520
 - Round 432 to 430
2. Add the rounded numbers:
 - $520 + 430 = 950$
3. Adjust:
 - Subtract the difference you added: $520 - 517 = 3$ and $430 - 432 = -2$
 - Total adjustment = $3 - 2 = 1$
4. Final answer:
 - $950 - 1 = 949$

Example 2: $1,251 - 867$

1. Round to nearest hundred:
 - Round 1,251 to 1,300
 - Round 867 to 900

2. Subtract the rounded numbers:

- $1,300 - 900 = 400$

3. Adjust:

- $1,300 - 1,251 = 49$ and $900 - 867 = 33$

- Total adjustment = $49 - 33 = 16$

4. Final answer:

- $400 - 16 = 384$

This strategy is especially effective when the numbers are fairly close to multiples of 10 or 100.

Strategy 3: Multiplying by 9 and Other Quick Tricks

Multiplying by 9 can sometimes be tricky for students, but there are some easy tricks to make this faster.

Why this Works:

By recognising certain patterns in multiplication, students can quickly find the answers without needing to go through long multiplication steps.

Multiply by 9 (using subtraction method)

Let's try 9×47 .

1. Multiply 47 by 10 (easy):

- $47 \times 10 = 470$

2. Subtract the original number (47) from this result:

- $470 - 47 = 423$

This method is handy when dealing with multiples of 9.

Strategy 4: Use the 'Split and Double' Method for Division

The "split and double" method is an excellent strategy for dividing by numbers like 4, 8, 12, and 16.

Why this Works:

This method allows students to split the division into easier, smaller steps and then double the result, which makes the calculations faster.

Example 1: Divide by 4

Let's divide 88 by 4.

1. Split the number:
 - Split 88 into 44 and 44.
2. Divide each part by 2 (since 4 is double of 2):
 - $44 \div 2 = 22$
 - $44 \div 2 = 22$

Example 2: Divide by 8

Let's divide 72 by 8.

1. Split 72 into 36 and 36.
2. Divide each part by 2:
 - $36 \div 2 = 18$
 - $36 \div 2 = 18$
3. Half
 - 18 halved is 9

This method works because splitting the number in half and then halving it again makes it simpler for students to grasp, especially when dividing by numbers like 4 and 8.

Strategy 5: Multiplying and Dividing by Powers of 10 (Quick Scale Adjustments)

Another crucial skill in mental arithmetic is being able to quickly multiply or divide by powers of 10 (like 10, 100, 1,000, etc.).

Why this Works:

When multiplying or dividing by powers of 10, all you need to do is shift the decimal point to the left or right, depending on whether you are dividing or multiplying.

Example 1: Multiply by 10, 100, and 1,000

Let's say we need to multiply 374 by 10, 100, and 1,000.

1. Multiply by 10:

- $374 \times 10 = 3,740$ (shift the decimal point one place to the right)
- 2. Multiply by 100:
 - $374 \times 100 = 37,400$ (shift the decimal point two places to the right)
- 3. Multiply by 1,000:
 - $374 \times 1,000 = 374,000$ (shift the decimal point three places to the right)

Example 2: Divide by 10, 100, and 1,000

Now let's divide 1,250 by 10, 100, and 1,000.

- 1. Divide by 10:
 - $1,250 \div 10 = 125$ (shift the decimal point one place to the left)
- 2. Divide by 100:
 - $1,250 \div 100 = 12.5$ (shift the decimal point two places to the left)
- 3. Divide by 1,000:
 - $1,250 \div 1,000 = 1.25$ (shift the decimal point three places to the left)

Strategy 6: Visualising Multiplication and Division

Visualization is a powerful tool in mental arithmetic. For example, let's say we have a multiplication problem like 12×6 .

Why this Works:

If students visualise the problem in terms of smaller, simpler numbers, they can often simplify it mentally without doing complex calculations.

Example 1: 12×6

- 1. Break 12 down:
 - $12 \times 6 = (10 \times 6) + (2 \times 6)$
- 2. Perform each operation:
 - $10 \times 6 = 60$
 - $2 \times 6 = 12$
- 3. Add the results:
 - $60 + 12 = 72$

This method works especially well when students need to multiply larger numbers. Encourage them to break down the numbers into simpler chunks.

Practice Questions for Mental Arithmetic

1. What is $1,749 + 623$?
2. What is 987×9 ?
3. What is $1,200 \div 8$?
4. What is $2,015 - 876$?
5. What is 547×25 ?

2. Word Problems (Multi-Step Problem Solving)

Word problems are a significant part of the GL Maths assessment, and they often involve real-world scenarios where students need to translate written language into mathematical equations. These problems can vary in difficulty, ranging from simple arithmetic to more complex multi-step reasoning tasks.

As an experienced tutor, I focus on teaching my students how to break down word problems, identify the relevant information, and systematically solve them. Below, I'll outline the best strategies I've developed to help my students become confident in tackling word problems.

Strategy 1: Identify Key Information (Highlighting)

The first and most important step in solving a word problem is to identify what information is important and what's extraneous. I always encourage students to underline or highlight the key pieces of information in each question.

Why this Works:

By highlighting key information, students can ensure they don't miss any critical details and help avoid getting bogged down by irrelevant data.

Example 1: Sarah's Books

Problem:

Sarah buys 3 books. Each book costs £8. How much money does she spend?

1. Key Information to Highlight:

- 3 books
- Each book costs £8

2. Equation:

- Total cost = Number of books $\times$ Price per book
- Total cost = 3×8

3. Solution:

- Total cost = £24

In this case, the process is straightforward, but highlighting helps reinforce focus. The student can easily see that they need to multiply the number of books by the price.

Tip:

When the problem is longer or more complex, make sure to highlight numbers, units of measurement (like kilograms or metres), and any words that tell you what operation to perform (like “total,” “difference,” “sum,” etc.).

Strategy 2: Break Down Multi-Step Problems (Work Step-by-Step)

Some word problems require multiple steps to reach the solution. It’s crucial to work through each part of the problem slowly and logically.

Why this Works:

Taking the problem one step at a time reduces the likelihood of mistakes and helps build confidence in tackling complex problems.

Example 2: Emma’s Party

Problem:

Emma buys 5 packs of balloons for a party. Each pack contains 12 balloons. After the party, 30 balloons are left. How many balloons were used at the party?

1. Step 1: Find the total number of balloons purchased.

- Total balloons = $5 \times 12 = 60$ balloons

2. Step 2: Subtract the remaining balloons.

- Balloons used = $60 - 30 = 30$ balloons

3. Solution:

- 30 balloons were used at the party.

This problem is solved by breaking it into two clear steps: first, multiplying to find the total number of balloons, then subtracting to find how many were used.

Tip:

For multi-step problems, encourage your child to write down each part of the solution, even if they think they know the answer. This helps avoid mistakes and ensures they are following the logical progression.

Strategy 3: Translate Words into Mathematical Equations

Many word problems require students to translate written words into mathematical symbols and equations. I often teach my students to look for keywords that signal mathematical operations.

Why this Works:

Identifying keywords and turning them into mathematical operations can make even the most complex word problems easier to solve.

Example 3: Tom's Savings

Problem:

Tom has £50. He spends £12 on a new book and £8 on a meal. How much money does he have left?

1. Step 1: Translate the word problem into an equation

- Start with £50, then subtract the costs.
- $50 - 12 - 8$

2. Step 2: Perform the subtraction.

- $50 - 12 = 38$
- $38 - 8 = £30$

3. Solution:

- £30 left after spending.

Here, the keywords are “spends” (which means subtraction) and “left” (indicating the result after subtraction).

Tip:

Students should get into the habit of identifying key phrases like:

- “Total” = addition
- “Difference” = subtraction
- “Per” = division
- “Times” or “of” = multiplication

Strategy 4: Use Estimation for Checks

Estimation can be a quick and effective way to ensure that your answer makes sense. Before diving into calculations, I encourage my students to roughly estimate the solution based on the problem's numbers.

Why this Works:

Estimation gives a ballpark figure, which can help students quickly spot errors in their calculation process.

Example 4: Anna's Cookies

Problem:

Anna bakes 32 cookies. She gives 5 cookies to each of her 6 friends. How many cookies does she have left?

1. Estimation:

- $5 \times 6 = 30$ cookies
- So, she will give away about 30 cookies.

2. Exact Calculation:

- Total cookies = 32
- Cookies given away = $5 \times 6 = 30$
- Cookies left = $32 - 30 = 2$ cookies

The estimate of 30 cookies is close to the actual result, so the final answer makes sense.

Tip:

Encourage your child to round numbers to the nearest 5 or 10 during estimation. This helps speed up the process while maintaining reasonable accuracy.

Strategy 5: Recognise Word Problem Patterns (Look for Similar Problems)

Word problems often follow a specific pattern or structure. Over the years, I've found that most of the problems in the GL Maths test fall into one of several categories.

Why this Works:

Recognizing the problem structure can save a lot of time and reduce anxiety for students.

Example 5: Buying Apples

Problem:

A packet of apples costs £2.50. How much would 4 packets of apples cost?

1. Recognize the problem type: This is a repeated addition problem.
 - The same price for each packet, so this is a case of multiplication.
2. Translate into a multiplication equation:
 - $4 \times £2.50 = £10.00$
3. Solution:
 - The total cost is £10.00.

Tip:

When your child encounters a word problem, ask them to classify it:

- Multiplication (repeated addition): “How much in total?” or “How many times?”
- Division (sharing equally): “How much per person?” or “How many in each group?”
- Subtraction (finding the difference): “How much more?” or “How much left?”

Recognizing the pattern early makes it easier to decide on the right operation.

Strategy 6: Check for Units of Measurement (Time, Money, Distance)

When solving word problems that involve time, money, or distance, always ensure that the units of measurement are consistent. I frequently remind my students to convert units when necessary.

Why this Works:

If units aren't consistent, the answer will likely be incorrect. Students should be adept at converting between units of measurement before proceeding with calculations.

Example 6: Travel Time

Problem:

A car travels 150 miles in 3 hours. How far will the car travel in 5 hours at the same speed?

1. Step 1: Identify the units
 - The speed is given in miles per hour (mph).
2. Step 2: Find the rate of travel

- $150 \text{ miles} \div 3 \text{ hours} = 50 \text{ miles per hour.}$
3. Step 3: Multiply by the time

- $50 \text{ miles/hour} \times 5 \text{ hours} = 250 \text{ miles}$

Tip:

When dealing with time, money, or distance, always check the units. Convert time to minutes if necessary, change currency if dealing with different denominations, or convert kilometers to miles if required.

Practice Questions for Word Problems:

1. Anna has 30 marbles. She gives 5 marbles to each of her 4 friends. How many marbles does she have left?
2. David is making a cake. He needs 200g of flour. He already has 150g. How much more flour does he need?
3. A train travels at 60 miles per hour. How far will it travel in 3.5 hours?
4. A box of chocolates costs £4.50. How much will 7 boxes cost?
5. Jane has 20 pencils. She gives 3 pencils to each of her friends. How many friends does she give pencils to?

Algebra (Understanding Patterns, Equations, and Variables)

Algebra in the GL Maths assessment often involves working with variables, simple equations, and patterns. The key to mastering algebraic questions is understanding that they are simply a set of rules to follow—once students learn how to manipulate numbers and variables, they'll find algebra much easier.

As an experienced 11+ tutor, I emphasise a step-by-step approach and use a combination of practical tips, practice, and logic to help students excel in this section.

Strategy 1: Understand the Concept of Variables

Algebra is essentially the study of relationships between numbers and unknowns, often represented by letters (variables). The first step is ensuring that students understand what a variable is and how to manipulate it.

Why this Works:

Many students struggle with algebra because they think the letters represent complex ideas. In reality, variables are just placeholders for numbers. Helping students become comfortable with this idea builds confidence.

Example 1: Simple Expressions

Consider the expression:

$$3x + 5 = 20$$

1. Identify the variable:
 - The variable is x .
2. Simplify the equation:
 - First, subtract 5 from both sides:
 $3x = 15$
3. Solve for x :
 - Divide both sides by 3:
 $x = 5$

Conclusion:

The value of x is 5.

Tip:

Encourage students to think of variables as "mystery numbers". They don't need to be scared of them! Just work through the equation like a puzzle.

Strategy 2: Balance Equations by Performing the Same Operation on Both Sides

In algebra, one of the most important principles is that whatever you do to one side of the equation, you must also do to the other side. This maintains the equality.

Why this Works:

The rule of keeping the equation balanced makes algebra feel more like a logical puzzle and less like a math mystery. This also helps avoid making errors by treating both sides equally.

Example 2: Solve for x

Solve:

$$5x - 3 = 22$$

1. Isolate the variable (x):
 - Add 3 to both sides to cancel the -3 :
 $5x = 25$
2. Solve for x :

- Divide both sides by 5:
 $x = 5$

Conclusion:

The value of x is 5.

Tip:

When working with more complex equations, always perform the same operation on both sides of the equation to preserve the equality. It's like keeping a balance on a scale—what you add to one side, you must add to the other!

Strategy 3: Work with Simple Linear Equations

A linear equation is an equation that involves only variables raised to the first power (no exponents). These are usually the types of equations that students encounter in the 11+ GL assessment.

Why this Works:

Linear equations are foundational in algebra. Once students can solve simple linear equations, they can apply the same principles to more complex ones.

Example 3: Solve for y

Solve:

$$4y + 6 = 30$$

1. Isolate the term with the variable ($4y$):

- Subtract 6 from both sides:
 $4y = 24$

2. Solve for y :

- Divide both sides by 4:
 $y = 6$

Conclusion:

The value of y is 6.

Tip:

For more complex equations, you might encounter multiple steps, but always follow the same basic process: isolate the variable, perform operations on both sides, and solve.

Strategy 4: Solve Problems Involving Unknowns on Both Sides

Occasionally, you will encounter equations with variables on both sides. In these cases, you need to move all the variables to one side and the constants to the other side.

Why this Works:

This strategy teaches students how to combine like terms and simplify equations before solving them, which is essential when tackling more difficult algebraic expressions.

Example 4: Solve for z

Solve:

$$2z + 3 = z + 8$$

1. Move all variables to one side:

○ Subtract z from both sides:

$$2z - z + 3 = 8$$

$$z + 3 = 8$$

2. Move constants to the other side:

○ Subtract 3 from both sides:

$$z = 5$$

Conclusion:

The value of z is 5.

Tip:

When variables appear on both sides of the equation, always aim to get all variables on one side (and constants on the other side) so you can isolate the variable and solve.

Strategy 5: Use Substitution to Solve Word Problems

Sometimes, word problems involve solving for variables by substituting known values into an equation. The idea is to replace the variable with numbers you already know.

Why this Works:

Substitution is a great technique for relating algebra to real-life problems. It allows students to build connections between abstract algebraic expressions and practical scenarios.

Example 5: Word Problem

Problem:

A bag contains only red and green apples. There are 15 apples in total. If the number of red apples is 4 more than twice the number of green apples, how many red apples are there?

1. Let x = the number of green apples.
Then, the number of red apples is $2x + 4$.
2. Set up the equation for total apples:
 - The total number of apples is 15:
 $x + (2x + 4) = 15$
3. Solve the equation:
 - Combine like terms:
 $3x + 4 = 15$
 - Subtract 4 from both sides:
 $3x = 11$
 - Divide by 3:
 $x = 11/3$, which doesn't give a whole number.

This means the problem is either inaccurate or has constraints we need to address. This teaches students to verify solutions and make sure their answers are reasonable.

Tip:

In real-world problems, make sure the solution fits logically—sometimes you might end up with an impossible answer that tells you the problem is flawed. Teach students to check for logical consistency.

Strategy 6: Understand and Recognise Sequences (Patterns)

Algebra often involves working with patterns or sequences. Recognizing these patterns is key in solving problems related to series, progressions, or patterns of change.

Why this Works:

Recognising patterns can help students solve algebraic problems more quickly, as they can apply a formula or a known method based on the pattern they see.

Example 6: Arithmetic Sequences

Problem:

The number of pencils in a box increases by 3 each day. The sequence begins with 2 pencils on Day 1. How many pencils are there on Day 5?

1. Recognise the pattern:
The sequence increases by 3 each day:
2, 5, 8, 11, 14 (Day 5)
2. Solution:
There are 14 pencils on Day 5.

Conclusion:

Recognising that this is a simple arithmetic sequence allows students to identify the pattern and find the solution without complex calculations.

Tip:

For sequences, always look for common differences (for arithmetic sequences) or common ratios (for geometric sequences). Identifying the type of sequence early makes solving them much faster.

1. Solve for x : $2x + 5 = 17$

Solution:

1. Isolate the variable (x):
Subtract 5 from both sides:
 $2x = 12$
2. Solve for x :
Divide both sides by 2:
 $x = 6$

Answer: $x = 6$

2. Solve for y : $4y - 9 = 19$

Solution:

1. Isolate the variable (y):

Add 9 to both sides:

$$4y = 28$$

2. Solve for y:

Divide both sides by 4:

$$y = 7$$

Answer: $y = 7$

3. Solve for x: $5(x + 3) = 30$

Solution:

1. Distribute the 5:

$$5x + 15 = 30$$

2. Isolate the variable (x):

Subtract 15 from both sides:

$$5x = 15$$

3. Solve for x:

Divide both sides by 5:

$$x = 3$$

Answer: $x = 3$

4. Solve for x: $2(x - 4) = 16$

Solution:

1. Distribute the 2:

$$2x - 8 = 16$$

2. Isolate the variable (x):

Add 8 to both sides:

$$2x = 24$$

3. Solve for x:

Divide both sides by 2:

$$x = 12$$

Answer: $x = 12$

5. Solve for x: $3x + 2 = 5x - 6$

Solution:

1. Move the variables to one side:
Subtract $3x$ from both sides:
 $2 = 2x - 6$
2. Move the constants to the other side:
Add 6 to both sides:
 $8 = 2x$
3. Solve for x:
Divide both sides by 2:
 $x = 4$

Answer: $x = 4$

6. Solve for x: $4x - 3 = 2(x + 4)$

Solution:

1. Distribute the 2 on the right-hand side:
 $4x - 3 = 2x + 8$
2. Move the variables to one side:
Subtract $2x$ from both sides:
 $2x - 3 = 8$
3. Move the constants to the other side:
Add 3 to both sides:
 $2x = 11$
4. Solve for x:
Divide both sides by 2:
 $x = 5.5$

Answer: $x = 5.5$

7. Word Problem: A number is decreased by 7 and the result is multiplied by 3 to give 30. What is the number?

Solution:

1. Let x be the unknown number.
The problem says the number is decreased by 7 and the result is multiplied by 3:
$$3(x - 7) = 30$$
2. Solve the equation:
Distribute the 3:
$$3x - 21 = 30$$
3. Isolate the variable:
Add 21 to both sides:
$$3x = 51$$
4. Solve for x :
Divide both sides by 3:
$$x = 17$$

Answer: The number is 17.

8. Word Problem: The sum of three consecutive numbers is 63. What are the numbers?

Solution:

1. Let the three consecutive numbers be x , $x+1$, and $x+2$.
The sum of these numbers is given as 63:
$$x + (x + 1) + (x + 2) = 63$$
2. Simplify the equation:
$$3x + 3 = 63$$
3. Solve for x :
Subtract 3 from both sides:
$$3x = 60$$

Divide both sides by 3:
$$x = 20$$
4. Find the three numbers:
The three consecutive numbers are 20, 21, and 22.

Answer: The numbers are 20, 21, and 22.

9. Solve for x : $7x + 9 = 4(x + 6)$

Solution:

1. Distribute the 4 on the right-hand side:
 $7x + 9 = 4x + 24$
2. Move the variables to one side:
Subtract $4x$ from both sides:
 $3x + 9 = 24$
3. Move the constants to the other side:
Subtract 9 from both sides:
 $3x = 15$
4. Solve for x :
Divide both sides by 3:
 $x = 5$

Answer: $x = 5$

10. Solve for x : $3(x + 5) = 2x + 30$

Solution:

1. Distribute the 3 on the left-hand side:
 $3x + 15 = 2x + 30$
2. Move the variables to one side:
Subtract $2x$ from both sides:
 $x + 15 = 30$
3. Move the constants to the other side:
Subtract 15 from both sides:
 $x = 15$

Answer: $x = 15$

11. Word Problem: A rectangle has a length that is 3 times its width. If the perimeter is 48 cm, find the length and width.

Solution:

1. Let the width be x .
The length is 3 times the width, so the length is $3x$.

2. Perimeter of a rectangle = $2(\text{length} + \text{width})$:

$$2(3x + x) = 48$$

3. Simplify the equation:

$$2(4x) = 48$$

$$8x = 48$$

4. Solve for x:

Divide both sides by 8:

$$x = 6$$

5. Find the length:

The length is $3x = 3(6) = 18$ cm.

Answer: The width is 6 cm, and the length is 18 cm.

12. Solve for x: $x/4 + 3 = 8$

Solution:

1. Isolate the term with x:

Subtract 3 from both sides:

$$x/4 = 5$$

2. Solve for x:

Multiply both sides by 4:

$$x = 20$$

Answer: $x = 20$

13. Word Problem: Jack has £50. He buys a t-shirt for £18 and a pair of shoes for £25. How much money does he have left?

Solution:

1. Identify the total spent:

Jack spends £18 on the t-shirt and £25 on the shoes.

$$\text{Total spent} = £18 + £25 = £43$$

2. Calculate the remaining money:

$$\text{Remaining money} = £50 - £43 = £7$$

Answer: Jack has £7 left.

14. Word Problem: A movie ticket costs £12. Sarah buys tickets for herself and 3 friends. How much does she spend in total?

Solution:

1. Number of tickets:
Sarah buys 4 tickets in total (one for herself and three for her friends).
2. Total cost:
 $\text{Total cost} = 4 \times £12 = £48$

Answer: Sarah spends £48 in total.

15. Solve for x: $2x - 5 = 3(x + 1)$

Solution:

1. Distribute the 3 on the right-hand side:
 $2x - 5 = 3x + 3$
2. Move the variables to one side:
Subtract $2x$ from both sides:
 $-5 = x + 3$
3. Move the constants to the other side:
Subtract 3 from both sides:
 $-8 = x$

Answer: $x = -8$

Place Value and Rounding

Understanding Place Value

Place value is the value of a digit based on its position within a number. For example, in the number 5,376, the 5 is in the thousands place, the 3 is in the hundreds place, the 7 is in the tens place, and the 6 is in the ones place.

Strategy 1: Break Down the Number into Place Values

This strategy involves understanding how to decompose a number based on its place values.

Why This Works:

Understanding how each digit in a number corresponds to a specific place value is essential. This strategy helps students visualise numbers and see their component parts.

Example: Decompose the Number 4,593

- 4 in the thousands place = 4,000
- 5 in the hundreds place = 500
- 9 in the tens place = 90
- 3 in the ones place = 3

So, $4,593 = 4,000 + 500 + 90 + 3$

Tip:

Encourage students to say the number out loud, breaking it into its components. For example, four thousand, five hundred and ninety-three. This will help reinforce the connection between the number and its place value.

Understanding Rounding

Rounding means adjusting a number to a nearby, more convenient number, often to a specific place value such as the nearest ten, hundred, or decimal place.

Strategy 2: Rounding to the Nearest Ten

When rounding to the nearest ten, focus on the ones place. If it's 5 or more, round up. If it's 4 or less, round down.

Why This Works:

Rounding to the nearest ten is one of the most basic but essential skills for number estimation. It helps students work quickly in tests and in real-life situations where exact numbers aren't necessary.

Example 1: Round 87 to the nearest ten.

- Look at the ones place: 7
- Since 7 is greater than 5, we round up.
- 87 rounds up to 90.

Example 2: Round 52 to the nearest ten.

- Look at the ones place: 2
- Since 2 is less than 5, we round down.
- 52 rounds down to 50.

Tip:

To help students get more comfortable with rounding, encourage them to think of number lines. For instance, 87 lies between 80 and 90, and rounding is simply a decision about which of these two values is closer to the number.

Strategy 3: Rounding to the Nearest Hundred

When rounding to the nearest hundred, focus on the tens place. If it's 5 or more, round up to the next hundred. If it's 4 or less, round down.

Why This Works:

Rounding to the nearest hundred is useful when dealing with large numbers and helps simplify calculations.

Example 1: Round 1,257 to the nearest hundred.

- Look at the tens place: 5
- Since 5 or greater, we round up.
- 1,257 rounds up to 1,300.

Example 2: Round 4,423 to the nearest hundred.

- Look at the tens place: 2
- Since 2 is less than 5, we round down.
- 4,423 rounds down to 4,400.

Tip:

When rounding to the nearest hundred, you can also use a number line to help visualise where the number lies between the two hundreds. For example, 1,257 is closer to 1,300 than it is to 1,200, so we round up.

Strategy 4: Rounding to the Nearest Thousand

When rounding to the nearest thousand, focus on the hundreds place. If it's 5 or more, round up. If it's 4 or less, round down.

Why This Works:

This strategy is important for dealing with very large numbers and helps students manage these more easily.

Example 1: Round 3,849 to the nearest thousand.

- Look at the hundreds place: 8

- Since 8 is greater than 5, we round up.
- 3,849 rounds up to 4,000.

Example 2: Round 6,352 to the nearest thousand.

- Look at the hundreds place: 3
- Since 3 is less than 5, we round down.
- 6,352 rounds down to 6,000.

Tip:

Make sure your child understands that when rounding to the nearest thousand, they are essentially finding the nearest "big" number—usually a round figure that is easier to work with mentally.

Strategy 5: Rounding Decimal Numbers

Rounding decimals is very similar to rounding whole numbers, but it requires focus on the specific decimal place.

Why This Works:

Rounding decimals is often required in real-life applications like money and measurement, making this a key skill to master.

Example 1: Round 3.74 to the nearest tenth.

- Look at the hundredths place: 4
- Since 4 is less than 5, we round down.
- 3.74 rounds to 3.7.

Example 2: Round 2.86 to the nearest tenth.

- Look at the hundredths place: 6
- Since 6 is greater than 5, we round up.
- 2.86 rounds to 2.9.

Tip:

For decimals, always emphasise place value. If rounding to the nearest tenth, check the hundredths place. For rounding to the nearest hundredth, check the thousandths place, and so on.

Additional Practice Questions

1. Round 563 to the nearest ten.
Answer: 560

2. Round 3,752 to the nearest hundred.
Answer: 3,800
3. Round 1,234 to the nearest thousand.
Answer: 1,000
4. Round 6.58 to the nearest tenth.
Answer: 6.6
5. Round 7,144 to the nearest hundred.
Answer: 7,100
6. Round 4.99 to the nearest whole number.
Answer: 5
7. Round 3.24 to the nearest tenth.
Answer: 3.2
8. Round 128 to the nearest ten.
Answer: 130
9. Round 2,438 to the nearest thousand.
Answer: 2,000
10. Round 9,749 to the nearest hundred.
Answer: 9,700

Key Tips for Success in Place Value and Rounding

- Understand the Basics: Make sure your child understands place value before jumping into rounding. This foundational concept helps them approach rounding with confidence.
- Use Number Lines: When in doubt, encourage your child to use a number line. It's a fantastic visual tool for seeing which value a number is closer to.
- Round Up or Down Based on the Neighbouring Digit: The general rule for rounding is simple: if the digit is 5 or greater, round up. If it's 4 or lower, round down.
- Practice, Practice, Practice: Rounding numbers quickly is a skill that comes with repetition. Give your child plenty of opportunities to practice both rounding and decomposing numbers into their place values

Type 4: Function/Number Machines

1. Understanding the Concept of Function Machines

A function machine is simply a series of operations that you perform on a number. Each operation will transform the input number in a certain way, and your task is to determine the output number.

Common operations you might see in function machines:

- Addition (+)
- Subtraction (-)
- Multiplication ($\times$)
- Division ($\div$)
- Combination of operations (e.g., multiply and then add)

For example:

A function machine might say:

“Multiply by 2, then add 5.”

If the input is 3:

- Multiply 3 by 2 = 6
- Add 5 = 11

Output: 11

Strategy 1: Understand the Steps

Before diving into the operations, read the problem carefully and break it down step by step. Identify what operations need to be performed in sequence, and be sure you're applying them in the right order.

2. Example Breakdown

Let's walk through a few step-by-step examples of function machines to illustrate the key strategies.

Example 1: Basic Function Machine

Function: "Multiply by 3, then subtract 5"

Input: 7

1. Multiply by 3:
 $7 \times 3 = 21$
2. Subtract 5:
 $21 - 5 = 16$

Output: 16

Answer: 16

Example 2: Reverse Function Machine

Sometimes you will need to figure out what the input was based on the output. This is called reverse reasoning.

Function: "Multiply by 4, then subtract 6"

Output: 14

1. Work backwards:
Start with the output: 14.
First, reverse the last operation (subtracting 6).
Add 6 to 14:
 $14 + 6 = 20$.
2. Reverse the multiplication:
Now divide by 4 to find the original input:
 $20 \div 4 = 5$.

Input: 5

Example 3: Multi-Step Function Machine

Function: "Multiply by 2, then add 10, then divide by 5."

Input: 25

1. Multiply by 2:
 $25 \times 2 = 50$
2. Add 10:
 $50 + 10 = 60$
3. Divide by 5:
 $60 \div 5 = 12$

Output: 12

3. Practice Questions

Question 1:

Function: "Multiply by 4, then subtract 8"

Input: 6

Solution:

1. $6 \times 4 = 24$

2. $24 - 8 = 16$

Answer: 16

Question 2:

Function: "Add 7, then divide by 3"

Input: 15

Solution:

1. $15 + 7 = 22$

2. $22 \div 3 = 7.33$ (This question has a decimal answer)

Answer: 7.33 (or approximately 7 if rounding)

Question 3:

Function: "Multiply by 5, then subtract 10"

Input: 10

Solution:

1. $10 \times 5 = 50$

2. $50 - 10 = 40$

Answer: 40

Question 4:

Function: "Subtract 12, then divide by 3"

Input: 42

Solution:

1. $42 - 12 = 30$

2. $30 \div 3 = 10$

Answer: 10

Question 5:

Function: "Multiply by 6, then divide by 3, then subtract 5"

Input: 9

Solution:

1. $9 \times 6 = 54$
2. $54 \div 3 = 18$
3. $18 - 5 = 13$

Answer: 13

4. Strategies for Solving Function Machines Effectively

Strategy 2: Work in Order of Operations

- Always follow the order of operations in the sequence that the machine gives you.
- If the machine gives multiple operations, follow them step-by-step.

For example:

In the machine "Multiply by 2, then subtract 6":

- Step 1: Multiply first
- Step 2: Subtract second

This order can't be reversed! A common mistake is performing the operations out of sequence.

Strategy 3: Reverse Operations for Input

If the problem provides an output and asks for the input, work backwards using the inverse of the operations. This requires a solid understanding of inverse operations:

- Inverse of addition (+) is subtraction (-)
- Inverse of multiplication ($\times$) is division ($\div$)
- Inverse of division ($\div$) is multiplication ($\times$)

For example, with the function machine:

"Add 5, then multiply by 4"

If the output is 36, you can work backwards to find the input.

1. Divide by 4: $36 \div 4 = 9$
2. Subtract 5: $9 - 5 = 4$

Input: 4

5. More Challenging Practice Questions

Question 6:

Function: "Add 3, multiply by 5, then divide by 2."

Input: 4

Solution:

1. $4 + 3 = 7$
2. $7 \times 5 = 35$
3. $35 \div 2 = 17.5$

Answer: 17.5

Question 7:

Function: "Divide by 4, subtract 3, then multiply by 8."

Input: 20

Solution:

1. $20 \div 4 = 5$
2. $5 - 3 = 2$
3. $2 \times 8 = 16$

Answer: 16

Question 8:

Function: "Multiply by 7, subtract 18, then divide by 3."

Input: 12

Solution:

1. $12 \times 7 = 84$
2. $84 - 18 = 66$
3. $66 \div 3 = 22$

Answer: 22

Question 9:

Function: "Add 8, then divide by 4, then multiply by 5."

Input: 12

Solution:

1. $12 + 8 = 20$
2. $20 \div 4 = 5$
3. $5 \times 5 = 25$

Answer: 25

Question 10:

Function: "Multiply by 6, subtract 12, then divide by 4."

Input: 14

Solution:

1. $14 \times 6 = 84$
2. $84 - 12 = 72$
3. $72 \div 4 = 18$

Answer: 18

6. Tips for Success in Function Machine Questions

1. Break it Down: Always carefully break the problem into smaller, manageable steps. Don't try to do too much at once.
2. Look for Patterns: Many function machine problems will involve a combination of similar operations, so look out for patterns and relationships between the operations.

\Understanding BODMAS (Brackets, Order, Divide, Multiply, Add, Subtract)

As an experienced 11+ tutor, one of the most essential concepts for the GL Maths assessment is mastering BODMAS(also known as PEMDAS in some places). This rule dictates the order of operations we need to follow when solving multi-step equations. Misapplying the order can lead to incorrect answers, which is why understanding BODMAS is crucial for success in both the 11+ and everyday maths!

Let's break this down step-by-step, using detailed examples, strategies, and practice questions to help your child understand how to use BODMAS in a variety of situations.

BODMAS Explained

BODMAS stands for:

- Brackets: First, solve any operations inside brackets.

- Order (or Exponents): This includes squares, cubes, or any other powers.
- Divide: Division comes before multiplication unless multiplication is directly inside brackets.
- Multiply: Multiplication comes after division, unless inside brackets.
- Add: After division and multiplication, perform addition.
- Subtract: Lastly, subtraction is done.

Strategy 1: Apply Brackets First (Even If It's Only One Set)

In any equation, the first thing to do is to solve the part inside brackets. If there are multiple sets of brackets, tackle each one sequentially, starting from the innermost.

For example:

Equation:

$$(5 + 3) \times 2$$

1. Step 1: Work inside the brackets first:
 $5 + 3 = 8$
2. Step 2: Multiply the result by 2:
 $8 \times 2 = 16$

Answer: 16

Strategy 2: Partition Method for Large Multiplications

Sometimes, you'll need to multiply large numbers that aren't easy to do mentally. In these cases, the Partition Method is very useful. This method breaks down the numbers into simpler components to make the multiplication manageable.

Example 1: Using the Partition Method in a BODMAS Question

Equation:

$$(77 \times 52) + (47 \times 25)$$

Step 1: Solve the first part of the equation inside the brackets:

$$77 \times 52$$

Use the Partition Method:

- Split 77 into 70 and 7, and split 52 into 50 and 2.
- Multiply 70 by 50:
 $70 \times 50 = 3500$

- Multiply 70 by 2:
 $70 \times 2 = 140$
- Multiply 7 by 50:
 $7 \times 50 = 350$
- Multiply 7 by 2:
 $7 \times 2 = 14$

Now add all of the products:

$$3500 + 140 + 350 + 14 = 4004$$

So, $77 \times 52 = 4004$.

Step 2: Solve the second part of the equation inside the brackets:

$$47 \times 25$$

Use the Partition Method again:

- Split 47 into 40 and 7, and split 25 into 20 and 5.
- Multiply 40 by 20:
 $40 \times 20 = 800$
- Multiply 40 by 5:
 $40 \times 5 = 200$
- Multiply 7 by 20:
 $7 \times 20 = 140$
- Multiply 7 by 5:
 $7 \times 5 = 35$

Now add all the products:

$$800 + 200 + 140 + 35 = 1175$$

So, $47 \times 25 = 1175$.

Step 3: Add the results from both parts together:

$$4004 + 1175 = 5179$$

Answer: 5179

Strategy 3: Multiplying and Dividing in Sequence

When the equation involves both multiplication and division, always work from left to right, as these two operations are of equal priority in BODMAS.

Example 2: $16 \div 4 \times 3$

1. Step 1: Start from the left side (division):
 $16 \div 4 = 4$
2. Step 2: Multiply the result by 3:
 $4 \times 3 = 12$

Answer: 12

Note: It's important to remember that multiplication and division are equal in BODMAS, so you must always work from left to right. The same rule applies if you have division and multiplication in the reverse order (e.g., $3 \times 5 \div 2$).

Strategy 4: Working with Addition and Subtraction

Addition and subtraction are handled last in BODMAS, but again, you must work from left to right. This rule is crucial when you have both addition and subtraction.

Example 3: $8 - 3 + 5$

1. Step 1: Start from the left side (subtraction):
 $8 - 3 = 5$
2. Step 2: Add 5 to the result:
 $5 + 5 = 10$

Answer: 10

Strategy 5: Applying the BODMAS Rule to Complex Equations

In more complex equations, you may need to apply multiple steps of BODMAS in sequence. It's important to break down each operation carefully to avoid mistakes.

Example 4: $(5 + 3) \times (8 - 2) \div 4$

1. Step 1: Solve inside the brackets first:
 $5 + 3 = 8$ and $8 - 2 = 6$
2. Step 2: Now multiply the two results:
 $8 \times 6 = 48$

3. Step 3: Divide the result by 4:
 $48 \div 4 = 12$

Answer: 12

Strategy 6: Applying the BODMAS Rule to Negative Numbers

Sometimes equations with negative numbers will come up, and you must handle them carefully according to BODMAS.

Example 5: $(3 + 4) \times (5 - 9) \div 2$

1. Step 1: Solve inside the brackets first:
 $3 + 4 = 7$ and $5 - 9 = -4$
2. Step 2: Multiply the results:
 $7 \times -4 = -28$
3. Step 3: Divide by 2:
 $-28 \div 2 = -14$

Answer: -14

Practice Questions

Question 1:

$$(6 + 4) \times (9 - 3)$$

Solution:

1. $6 + 4 = 10$
2. $9 - 3 = 6$
3. $10 \times 6 = 60$

Answer: 60

Question 2:

$$(15 + 5) \div 5 + 6$$

Solution:

1. $15 + 5 = 20$
2. $20 \div 5 = 4$

3. $4 + 6 = 10$

Answer: 10

Question 3:

$(8 \times 4) - (5 + 3)$

Solution:

1. $8 \times 4 = 32$

2. $5 + 3 = 8$

3. $32 - 8 = 24$

Answer: 24

Key Tips for Success with BODMAS:

1. Always start with brackets. This is the first rule in BODMAS and should be your starting point for any equation.
2. Work from left to right for multiplication and division, and addition and subtraction.
3. Remember negative numbers—handle them carefully, especially when multiplying and dividing.
4. Break down complex multiplications into smaller steps (using the partition method) to make them easier to solve.
5. Practice with mixed operations—the more complex the problem, the more you'll need to rely on BODMAS to guide your work.
6. Don't rush. BODMAS can feel tricky, but with practice and patience, your child will develop the confidence to solve any equation correctly.

Understanding Inverse Operations

Inverse operations are the opposite of the given mathematical operation, and they are key to solving equations with missing elements. In simple terms, "inverse" means "opposite", so if we're given an operation like multiplication, we can reverse it by dividing, or if addition is involved, we reverse it by subtracting.

In GL Maths questions, inverse operations are often used to help find unknown numbers in equations. Mastering inverse operations is crucial for solving a wide variety of questions on the 11+ exam.

Inverse Operations in Basic Terms

1. Addition ↔ Subtraction

If the equation involves addition, the inverse operation is subtraction.

Example:

If $x + 6 = 10$, subtract 6 from both sides to solve for x :

$$x + 6 - 6 = 10 - 6 \text{ so } x = 4$$

2. Multiplication ↔ Division

If the equation involves multiplication, the inverse operation is division.

Example:

If $3 \times x = 15$, divide both sides by 3 to solve for x :

$$3 \times x \div 3 = 15 \div 3 = 5$$

3. Division ↔ Multiplication

If the equation involves division, the inverse operation is multiplication.

Example:

If $x \div 4 = 5$, multiply both sides by 4 to solve for x :

$$x = 5 \times 4 \text{ so } x = 20$$

4. Square ↔ Square Root

If the equation involves squaring a number, the inverse is finding the square root of the number.

Example:

If $x^2 = 16$, find the square root of both sides:

$$x = \sqrt{16} = 4$$

Inverse Operations in Equations with Fractions

When fractions are involved, the concept of inverse operations still applies, but with an additional step to handle the fractions correctly.

Example 1: Inverse Operations with Fractions

Equation:

$$(2/3) + x = (5/6). \text{ What is } x?$$

Step-by-Step Solution:

1. Identify the operation: The equation is adding x to $(2/3)$, so the inverse of addition is subtraction.
2. Apply the inverse operation: To find x , subtract $(2/3)$ from both sides:
scss

$$x = (5/6) - (2/3)$$

3. Find a common denominator: The denominators are 6 and 3, so we convert $(2/3)$ to $(4/6)$ (since $3 \times 2 = 6$).

$$x = (5/6) - (4/6)$$

4. Simplify:
 $x = (5 - 4)/6$

5. $x = 1/6$

Answer: $x = 1/6$

Example 2: Inverse Operations with Mixed Numbers

If mixed numbers are involved, you will need to convert them into improper fractions before applying inverse operations.

Equation:

$$2 \frac{4}{5} + x = 3 \frac{2}{15}. \text{ What is } x?$$

Step-by-Step Solution:

1. Convert mixed numbers to improper fractions:
 - $2 \frac{4}{5}$ is equal to $(14/5)$ (since $2 \times 5 + 4 = 14$).
 - $3 \frac{2}{15}$ is equal to $(47/15)$ (since $3 \times 15 + 2 = 47$).
2. Rewrite the equation:
 $(14/5) + x = (47/15)$
3. Find a common denominator: The denominators are 5 and 15, so we can multiply $(14/5)$ by 3 to make it $(42/15)$.

- 4 • Apply the inverse operation: To solve for x , subtract $(42/15)$ from both sides:

$$x = (47/15) - (42/15)$$

Simplify:

$$x = (47 - 42)/15$$

5 • $x = 5/15$

6 • $x = 1/3$

Answer: $x = 1/3$

Example 3: Inverse Operations with Negative Numbers

Inverse operations also apply to equations involving negative numbers. Here's an example:

Equation:

$$x - 7 = -3. \text{ What is } x?$$

Step-by-Step Solution:

- 1 • Identify the operation: The equation involves subtraction, so the inverse operation is addition.
- 2 • Apply the inverse operation: To solve for x , add 7 to both sides:

$$x - 7 + 7 = -3 + 7$$

Answer: $x = 4$

Practice Questions

Question 1:

$$x + 5 = 12. \text{ What is } x?$$

Solution:

- 1 • Subtract 5 from both sides:

$$x = 12 - 5$$

Answer: $x = 7$

Question 2:

$4 \times x = 36$. What is x ?

Solution:

1. Divide both sides by 4:

$$x = 36 \div 4$$

Answer: $x = 9$

Question 3:

$x \div 3 = 9$. What is x ?

Solution:

1. Multiply both sides by 3:

$$x = 9 \times 3$$

2. $x = 27$

Answer: $x = 27$

Key Tips for Solving Inverse Operations

1. Always isolate the unknown by performing the inverse operation on both sides of the equation.
2. Use addition/subtraction first to simplify the equation if it involves those operations.
3. Work with fractions carefully, finding a common denominator when necessary.
4. Check your answer by substituting the value of x back into the original equation.
5. Practice with different operations—inverse operations can appear in many forms, so the more practice your child gets, the more comfortable they will become.

Factors and Multiples: Divisibility Rules & Prime Factors

Understanding factors, multiples, and prime factors is crucial for solving many types of questions in the 11+ GL Maths exam. These concepts are used to simplify problems, check divisibility, and break down numbers into their prime components. Let's break down each of these concepts in detail.

Factors

A factor of a number is any whole number that divides exactly into that number, leaving no remainder. For example, 1 and 2 are factors of 4 because:

- $4 \div 1 = 4$
- $4 \div 2 = 2$

So, 1 and 2 are factors of 4. Factors always come in pairs, like 1 and 4 or 2 and 2 for the number 4.

Prime Factors

A prime factor is a factor of a number that is also a prime number. A prime number is a number greater than 1 that has only two factors: 1 and itself.

For example:

- 2, 3, 5, 7, 11, and 13 are prime numbers.
- 30 has the factors 1, 2, 3, 5, 6, 10, 15, and 30. Among these, the prime factors of 30 are 2, 3, and 5 because they are prime numbers.

To break down a number into prime factors, use factor trees or division. For example:

For 30:

- Divide 30 by 2 (the smallest prime number):
 $30 \div 2 = 15$
- Now divide 15 by 3 (the next smallest prime):
 $15 \div 3 = 5$ (and 5 is a prime number).

So, the prime factors of 30 are:

$$2 \times 3 \times 5$$

Multiples

A multiple of a number is the product of that number and any other whole number.

For example, the multiples of 2 are:

2, 4, 6, 8, 10, 12, 14, 16, 18, etc.

In the same way, the multiples of 13 are:

13, 26, 39, 52, 65, 78, 91, 104, etc.

Divisibility Rules

Divisibility rules help us quickly determine if a number is divisible by another number, without performing the actual division. Below are some key divisibility rules:

- Divisible by 2: The number must end in an even digit (0, 2, 4, 6, or 8).
- Divisible by 3: The sum of the digits must be divisible by 3.
- Divisible by 5: The number must end in 0 or 5.
- Divisible by 10: The number must end in 0.

For more complex numbers, such as 13, testing divisibility often involves performing some division, but there are clever strategies you can use to check multiples quickly, as demonstrated in the example below.

Example Question: Finding Multiples of 13

In this question, we are asked to find which of the five options is a multiple of 13. This means we need to find which number can be divided by 13 without leaving a remainder.

Options:

- A) 281
- B) 130
- C) 106
- D) 208
- E) 91

Step-by-Step Solution:

1. Start with simple multiples of 13:

It's often helpful to know a few multiples of 13 off the top of your head. For example:

- $13 \times 10 = 130$
- $13 \times 20 = 260$
- $13 \times 30 = 390$

2. Approximate nearest multiples:

We can see that 130 is a multiple of 13, so we can use it as a reference to check which of the options are divisible by 13.

- Option A: 281
 - $13 \times 20 = 260$ (close, but not 281).
 - $13 \times 21 = 273$ (still not 281).
 - $13 \times 22 = 286$ (not 281).
 - So, 281 is not divisible by 13.
- Option B: 130
 - $13 \times 10 = 130$.
 - 130 is divisible by 13, so this is a valid answer.

- Option C: 106
 - $13 \times 8 = 104$ (close, but not 106).
 - $13 \times 9 = 117$ (again, not 106).
 - So, 106 is not divisible by 13.
- Option D: 208
 - $13 \times 16 = 208$.
 - 208 is divisible by 13, so this is another valid answer.
- Option E: 91
 - $13 \times 7 = 91$.
 - 91 is divisible by 13, so this is another valid answer.

Conclusion:

The correct options are B (130), D (208), and E (91).

Practice Question 1: Prime Factorization

Question: What are the prime factors of 60?

Solution:

1. Divide 60 by the smallest prime number, 2: $60 \div 2 = 30$
2. Divide 30 by 2 again: $30 \div 2 = 15$
3. Divide 15 by 3 (next smallest prime number): $15 \div 3 = 5$
4. Finally, 5 is a prime number, so we stop here.

Prime factors of 60:

$$2 \times 2 \times 3 \times 5$$

Practice Question 2: Divisibility by 3

Question: Is 864 divisible by 3?

Solution:

1. Add the digits of 864: $8+6+4=18$
2. Since 18 is divisible by 3, 864 is divisible by 3.

Answer: Yes, 864 is divisible by 3.

Practice Question 3: Finding Factors

Question: Which of the following is not a factor of 36?

- A) 6
- B) 9
- C) 12
- D) 15

Solution:

- The factors of 36 are: 1, 2, 3, 4, 6, 9, 12, 18, 36.
- 15 is not on this list, so the correct answer is D (15).

Highest Common Factors (HCF) and Lowest Common Multiples (LCM)

When dealing with questions in the 11+ GL Maths exam, understanding how to find Highest Common Factors (HCF) and Lowest Common Multiples (LCM) is crucial. These concepts not only help in simplifying fractions, but also play a big role in solving various types of word problems and multiple-choice questions. Let's break down these concepts with clear strategies and examples.

Highest Common Factor (HCF)

The Highest Common Factor (HCF) is the largest number that divides exactly into two or more numbers. In other words, it's the greatest factor that both numbers share.

How to Find the HCF:

1. List the Factors: Start by listing the factors of each number.
2. Identify Common Factors: Find the numbers that appear in both lists.
3. Choose the Largest Common Factor: The highest of these common factors is the HCF.

Example 1: Finding the HCF of 12 and 16

1. List the factors of 12:
1, 2, 3, 4, 6, 12
2. List the factors of 16:
1, 2, 4, 8, 16
3. Identify the common factors:
1, 2, 4
4. Choose the largest common factor:
4 is the HCF of 12 and 16.

So, $\text{HCF}(12, 16) = 4$.

Lowest Common Multiple (LCM)

The Lowest Common Multiple (LCM) is the smallest multiple that both numbers share. In simpler terms, it's the first number that appears in both of their times tables.

How to Find the LCM:

1. List the multiples of each number.
2. Identify the common multiples.
3. Choose the smallest common multiple.

Example 2: Finding the LCM of 7 and 13

1. List the multiples of 7:
7, 14, 21, 28, 35, 42, 49, 56, 63, 70, ...
2. List the multiples of 13:
13, 26, 39, 52, 65, 78, 91, 104, 117, ...
3. Identify the common multiples:
The smallest common multiple is 91.

So, $\text{LCM}(7, 13) = 91$.

Using HCF and LCM in the 11+ Exam

For multiple-choice questions, it's often efficient to use the options provided to work backwards. This strategy can save time when dealing with higher numbers.

Example 3: Finding the LCM from the Options (Multiple Choice)

Let's say we are given a question that asks us to find the LCM of 7 and 13, and the options are as follows:

- A) 26
- B) 182
- C) 65
- D) 143
- E) 247

We know that $7 \times 13 = 91$, but let's use the options to narrow it down:

1. Option A: 26
26 is not a multiple of 7 or 13.

2. Option B: 182
182 is not divisible by 7 or 13.
3. Option C: 65
65 is not divisible by 7 or 13.
4. Option D: 143
143 is not divisible by 7 or 13.
5. Option E: 247
247 is divisible by both 7 and 13, but it's not the smallest common multiple.

So, we know that 91 is the LCM of 7 and 13, which means the correct answer is B: 182.

Example 4: Highest Common Factor in Word Problems

Question:

Find the HCF of 24 and 36.

Solution:

1. List the factors of 24:
1, 2, 3, 4, 6, 8, 12, 24
2. List the factors of 36:
1, 2, 3, 4, 6, 9, 12, 18, 36
3. Identify the common factors:
1, 2, 3, 4, 6, 12
4. Choose the largest common factor:
12 is the HCF.

Answer: $\text{HCF}(24, 36) = 12$

Practice Questions

Practice Question 1: LCM of 6 and 8

What is the LCM of 6 and 8?

1. List the multiples of 6:
6, 12, 18, 24, 30, 36, ...

2. List the multiples of 8:
8, 16, 24, 32, 40, ...
3. Identify the common multiples:
The smallest common multiple is 24.

Answer: $\text{LCM}(6, 8) = 24$

Practice Question 2: HCF of 18 and 24

What is the HCF of 18 and 24?

1. List the factors of 18:
1, 2, 3, 6, 9, 18
2. List the factors of 24:
1, 2, 3, 4, 6, 8, 12, 24
3. Identify the common factors:
1, 2, 3, 6
4. Choose the largest common factor:
6 is the HCF.

Answer: $\text{HCF}(18, 24) = 6$

Practice Question 3: LCM of 10 and 15

What is the LCM of 10 and 15?

1. List the multiples of 10:
10, 20, 30, 40, 50, 60, ...
2. List the multiples of 15:
15, 30, 45, 60, 75, ...
3. Identify the common multiples:
The smallest common multiple is 30.

Answer: $\text{LCM}(10, 15) = 30$

Key Strategies for HCF and LCM Questions

1. For HCF:

- Always list the factors of the numbers and find the largest common factor.
- If you get stuck, use prime factorization to break the numbers down into their prime components and identify common primes.

2. For LCM:

- Use the list of multiples and identify the smallest common multiple.

Number Sequences & Number Patterns

Understanding number sequences and patterns is a key skill that comes up frequently in the 11+ GL Maths exam. These questions test your child's ability to recognise trends, make predictions, and apply mathematical reasoning to find the next number or identify a rule governing a sequence.

What Are Number Sequences?

A number sequence is simply a list of numbers arranged in a specific order. Often, the numbers in a sequence follow a certain pattern, whether they increase or decrease by a fixed amount, or change according to a more complex rule.

For example:

- 1, 3, 5, 7, 9... is a sequence where each number increases by 2.
- 10, 20, 30, 40, 50... is a sequence where each number increases by 10.

Recognising these patterns quickly will allow your child to answer questions faster and more confidently. Let's go over a variety of sequence types you might encounter in the exam.

Type 1: Arithmetic Sequences

An arithmetic sequence is one where each number in the sequence is obtained by adding the same number to the previous number.

Example 1: Simple Addition Pattern

Question:

What is the next number in the sequence:

2, 5, 8, 11, ___?

Solution:

- Start by identifying the pattern. Here, we see that each number increases by 3.

- The next number is simply $11 + 3 = 14$.

Answer: 14

Type 2: Sequences Involving Multiplication

In some sequences, each term is obtained by multiplying the previous term by a constant. These are called geometric sequences.

Example 2: Simple Multiplication Pattern

Question:

What is the next number in the sequence:

2, 6, 18, 54, ___?

Solution:

- The rule is that each number is multiplied by 3 to get the next term.
- To find the next number, multiply 54 by 3:
 $54 \times 3 = 162$

Answer: 162

Type 3: Alternating Sequences

Some sequences alternate between two or more different patterns. These are trickier, but with careful observation, they can be solved easily.

Example 3: Alternating Pattern

Question:

What is the next number in the sequence:

2, 4, 6, 10, 12, ___?

Solution:

- First, observe that the odd-numbered terms increase by 2 (2, 6, 12), and the even-numbered terms increase by 4 (4, 10).
- The next odd-numbered term is $12 + 2 = 14$.

Answer: 14

Type 4: Sequences with Addition and Subtraction

Some sequences alternate between adding and subtracting. This requires careful attention to the alternating pattern.

Example 4: Addition and Subtraction Pattern

Question:

What is the next number in the sequence:

10, 15, 12, 17, ___?

Solution:

- The pattern alternates between adding 5 and subtracting 3.
- From 17, subtract 3:
 $17 - 3 = 14$

Answer: 14

Type 5: Sequences Involving Decimals

Number sequences don't always consist of whole numbers. Your child may encounter sequences involving decimals or money amounts.

Example 5: Decimal Addition Pattern

Question:

What is the next number in the sequence:

0.5, 1.0, 1.5, 2.0, ___?

Solution:

- The pattern increases by 0.5 each time.
- To find the next term, add 0.5 to 2.0:
 $2.0 + 0.5 = 2.5$

Answer: 2.5

Type 6: Money Sequences

Money-related number sequences are often featured in the 11+ exam, as it's a practical application of sequences in everyday life, like budgeting or calculating costs.

Example 6: Adding Money Amounts

Question:

Kevin spends £0.99 on a vanilla cone, £1.50 on a hot dog, and Jason spends £1.85 on a burger. How much money do they spend altogether?

Solution:

- Simply add the amounts together:
 $£0.99 + £1.50 + £1.85 = £4.34$

Answer: £4.34

Type 7: Missing Numbers in Sequences

Sometimes, a sequence may have a missing number, and you need to identify the rule and work out what the missing number is.

Example 7: Find the Missing Term

Question:

What is the missing number in the sequence:

3, 6, __, 12, 15?

Solution:

- The sequence increases by 3 each time (3, 6, 9, 12, 15).
- Therefore, the missing number is 9.

Answer: 9

Practice Questions

Here are some practice questions to help your child master the skill of recognising and completing number sequences:

Practice Question 1: Arithmetic Sequence

What is the next number in the sequence:

5, 8, 11, 14, __?

Solution:

- The sequence increases by 3.
- So, $14 + 3 = 17$.

Answer: 17

Practice Question 2: Geometric Sequence

What is the next number in the sequence:

4, 8, 16, 32, __?

Solution:

- The sequence multiplies by 2 each time.
- So, $32 \times 2 = 64$.

Answer: 64

Practice Question 3: Alternating Addition/Subtraction Sequence

What is the next number in the sequence:

20, 15, 20, 15, ___?

Solution:

- The sequence alternates between subtracting 5 and adding 5.
- The next term will be $15 + 5 = 20$.

Answer: 20

Practice Question 4: Decimal Sequence

What is the next number in the sequence:

1.5, 2.0, 2.5, 3.0, ___?

Solution:

- The sequence increases by 0.5 each time.
- The next term will be $3.0 + 0.5 = 3.5$.

Answer: 3.5

Practice Question 5: Money Sequence

If Kevin spends £2.45 on a sandwich, £3.60 on a drink, and Jason spends £5.75 on a dessert, how much do they spend altogether?

Solution:

- Add the amounts together:
 $£2.45 + £3.60 + £5.75 = £11.80$

Answer: £11.80

Strategies for Success in Number Sequences

1. Look for Common Differences:

- If the numbers are increasing or decreasing by a consistent amount, it's an arithmetic sequence.
- 2. Check for Multiplication:
 - If the numbers are being multiplied, it's a geometric sequence.
- 3. Identify Repeated Patterns:
 - Look out for patterns that repeat, such as alternating additions and subtractions.

Type 10 - Money & Cost Calculations

Money and cost calculations are another important skill for the 11+ GL Maths exam. These questions typically require your child to work with decimal numbers and apply basic operations like addition, subtraction, multiplication, and division to determine the total cost of various items or services. Mastering money and cost calculations will not only help your child with the 11+ but also in everyday situations, such as budgeting for lunch at school or buying items with pocket money.

What is Money & Cost Calculation?

Money and cost calculation questions involve working out the total cost of items, determining how much change is left after a purchase, or solving word problems related to prices and discounts. These questions usually involve decimals because prices often include pence.

Strategies for Solving Money & Cost Calculation Problems

1. Convert to Same Units
Ensure all amounts are in the same units. Typically, money is expressed in pounds (£) and pence (p), but sometimes you may need to convert between the two.
For example, £1.50 is the same as 150p.
2. Double-Check Decimal Places
Ensure you're consistent with decimal places when adding or subtracting money. It's crucial to align the decimal points to avoid errors.
3. Round for Estimation
Rounding the numbers to the nearest pound or 10p can help make quick estimations and checks to see if the final answer is reasonable.
4. Use Addition and Subtraction
Most money problems require adding up totals or finding the difference between amounts, such as calculating change.

Example 1: Adding Costs of Items

Question:

Kevin spends £0.99 on a vanilla cone and £1.50 on a hot dog. Jason spends £1.85 on a burger. How much do they spend altogether?

Solution:

To find the total amount they spend, we simply add up the three amounts:

- $£0.99 + £1.50 + £1.85 = £4.34$

Answer: £4.34

Example 2: Finding the Change After a Purchase

Question:

Alex buys a sandwich for £3.50 and a drink for £1.20. He gives the cashier £10. How much change does he receive?

Solution:

- First, calculate the total cost of the items:
 $£3.50 + £1.20 = £4.70$
- Then, subtract this total from £10 to find the change:
 $£10.00 - £4.70 = £5.30$

Answer: £5.30

Example 3: Calculating the Total Cost of Multiple Items

Question:

A pencil costs £0.75, an eraser costs £0.50, and a ruler costs £1.20. How much do all three items cost?

Solution:

- Add the prices of all three items:
 $£0.75 + £0.50 + £1.20 = £2.45$

Answer: £2.45

Example 4: Working Out the Cost of Multiple Items with Discounts

Question:

A jacket costs £40.00, but there is a 15% discount. How much does the jacket cost after the discount?

Solution:

- First, calculate 15% of £40.00:
 $15\% \text{ of } £40.00 = 0.15 \times 40 = £6.00$
- Subtract the discount from the original price:
 $£40.00 - £6.00 = £34.00$

Answer: £34.00

Example 5: Calculating Total Cost with VAT (Value Added Tax)

Question:

A toy costs £12.00. The VAT is 20%. How much will the total cost be including VAT?

Solution:

- First, calculate the VAT:
 $20\% \text{ of } £12.00 = 0.20 \times £12.00 = £2.40$
- Add the VAT to the original price:
 $£12.00 + £2.40 = £14.40$

Answer: £14.40

Example 6: Solving a Word Problem Involving Money

Question:

Mia wants to buy 3 pens that cost £1.25 each. She also wants to buy a notebook that costs £3.00. How much money does Mia need to bring if she has £5.50?

Solution:

- Calculate the total cost of the pens:
 $3 \times £1.25 = £3.75$
- Add the cost of the notebook:
 $£3.75 + £3.00 = £6.75$
- Now, compare this total with the amount Mia has:
 $£6.75 \text{ (total cost)} - £5.50 \text{ (amount Mia has)} = £1.25$
- Mia needs £1.25 more to cover the total cost.

Answer: Mia needs £1.25 more.

Example 7: Finding the Unit Price

Question:

A pack of 5 pens costs £2.50. What is the price of each pen?

Solution:

- Divide the total cost by the number of pens:
 $£2.50 \div 5 = £0.50$
- Each pen costs £0.50.

Answer: £0.50 per pen

Example 8: Money Conversion between Pounds and Pence

Question:

A toy costs £2.40. How much is this in pence?

Solution:

- To convert pounds to pence, multiply by 100:
 $£2.40 \times 100 = 240$ pence

Answer: 240p

Practice Questions

Here are some additional practice questions to help your child become proficient in money and cost calculations:

Practice Question 1: Adding Money

Question:

Emma buys a cake for £3.50, a drink for £1.75, and a bag of crisps for £0.99. How much does she spend in total?

Solution:

- Add the amounts:
 $£3.50 + £1.75 + £0.99 = £6.24$

Answer: £6.24

Practice Question 2: Finding Change

Question:

If Luke spends £6.80 on a book and pays with a £10 note, how much change does he get?

Solution:

- Subtract the total cost from £10:
 $£10 - £6.80 = £3.20$

Answer: £3.20

Practice Question 3: Calculating Total Cost with Discount

Question:

A pair of shoes costs £50.00, but there is a 20% discount. How much will the shoes cost after the discount?

Solution:

- Calculate 20% of £50.00:
 $20\% \text{ of } £50 = £10.00$
- Subtract the discount:
 $£50.00 - £10.00 = £40.00$

Answer: £40.00

Practice Question 4: Adding Money with Decimals

Question:

Sarah buys a pencil for £1.10, an eraser for £0.80, and a sharpener for £1.40. How much does she spend in total?

Solution:

- Add the prices:
 $£1.10 + £0.80 + £1.40 = £3.30$

Answer: £3.30

Practice Question 5: Finding Cost Including VAT

Question:

A phone costs £200.00, and the VAT is 18%. What is the total cost including VAT?

Solution:

- Calculate the VAT:
 $18\% \text{ of } £200.00 = £36.00$
- Add the VAT to the original price:
 $£200.00 + £36.00 = £236.00$

Answer: £236.00

Fractions

Fractions can seem intimidating to some students, but the key is practice and understanding. Once your child grasps the fundamental operations with fractions, they will find the questions easier to approach, and with regular practice, they will develop the confidence to tackle them without hesitation.

In the 11+ GL Maths exam, a wide range of questions involving fractions could appear, including:

- Adding, subtracting, multiplying, and dividing fractions
- Converting fractions to decimals or percentages
- Ordering fractions
- Simplifying fractions

By building a strong understanding of fractions, your child will be prepared for all of these types of questions. Let's go through the key concepts in more detail, and I'll give you step-by-step strategies to master each one.

Understanding Fractions: Key Concepts

1. Fraction Operations:

- **Addition and Subtraction of Fractions:** When adding or subtracting fractions, it's crucial that the fractions have a common denominator. If they don't, find the least common denominator (LCD) before performing the operation.
- **Multiplication of Fractions:** Multiply the numerators and denominators separately. The rule here is simple:
 $\text{Numerator} \times \text{Numerator} / \text{Denominator} \times \text{Denominator}$.
- **Division of Fractions:** When dividing fractions, you multiply the first fraction by the reciprocal of the second fraction (invert the second fraction).

Strategy for Solving Fraction Problems

Step 1: Understanding the Problem

When given a fraction problem, first decide if the question is asking to add, subtract, multiply, or divide fractions. Often, the key to solving fraction problems quickly is determining what operation is required.

Step 2: Convert Mixed Numbers and Improper Fractions

In some cases, your child may be asked to work with mixed numbers (like $3 \frac{1}{2}$) or improper fractions (like $\frac{7}{3}$). It's useful to know how to convert between the two:

- To convert a mixed number to an improper fraction, multiply the whole number by the denominator, then add the numerator.

Example: Convert $2 \frac{3}{4}$ to an improper fraction.

$$2 \times 4 + 3 = 8 + 3 = 11 \rightarrow \frac{11}{4}.$$

- To convert an improper fraction to a mixed number, divide the numerator by the denominator and express the remainder as a fraction.

Example: Convert $\frac{11}{4}$ to a mixed number.

$$11 \div 4 = 2 \text{ remainder } 3 \rightarrow 2 \frac{3}{4}.$$

Step 3: Simplify Fractions

Once you perform the operation, check if you can simplify the resulting fraction.

Simplifying involves finding the greatest common divisor (GCD) of the numerator and denominator, and dividing both by it.

Example 1: Multiplying Fractions

Question:

What is $\frac{3}{5} \times \frac{4}{7}$?

Solution:

- Multiply the numerators:
 $3 \times 4 = 12$
- Multiply the denominators:
 $5 \times 7 = 35$
- The result is:
 $\frac{12}{35}$

There is no common factor, so it's already simplified.

Answer: $\frac{12}{35}$

Example 2: Dividing Fractions

Question:

What is $\frac{3}{4} \div \frac{2}{5}$?

Solution:

- To divide by a fraction, multiply by the reciprocal of the second fraction. The reciprocal of $\frac{2}{5}$ is $\frac{5}{2}$.

- So we now have $\frac{3}{4} \times \frac{5}{2}$.
- Multiply the numerators:
 $3 \times 5 = 15$
- Multiply the denominators:
 $4 \times 2 = 8$
- The result is:
 $\frac{15}{8}$, which is an improper fraction. To convert it into a mixed number:
 $15 \div 8 = 1 \text{ remainder } 7 \rightarrow 1 \frac{7}{8}$.

Answer: $\frac{15}{8}$ or $1 \frac{7}{8}$

Example 3: Adding Fractions

Question:

What is $\frac{3}{8} + \frac{5}{8}$?

Solution:

- Since the denominators are the same (both 8), we simply add the numerators:
 $3 + 5 = 8$
- The result is:
 $\frac{8}{8} = 1$

Answer: 1

Example 4: Subtracting Fractions

Question:

What is $\frac{7}{10} - \frac{3}{10}$?

Solution:

- Since the denominators are the same (both 10), we simply subtract the numerators:
 $7 - 3 = 4$
- The result is:
 $\frac{4}{10}$, which simplifies to $\frac{2}{5}$ (divide both the numerator and denominator by 2).

Answer: $\frac{2}{5}$

Example 5: Converting Fractions to Decimals

Question:

Convert $\frac{7}{8}$ to a decimal.

Solution:

- Divide 7 by 8:
 $7 \div 8 = 0.875$.

Answer: 0.875

Example 6: Converting Fractions to Percentages

Question:

Convert $\frac{3}{5}$ to a percentage.

Solution:

- First, convert the fraction to a decimal:
 $3 \div 5 = 0.6$
- Then, multiply the decimal by 100 to get the percentage:
 $0.6 \times 100 = 60\%$.

Answer: 60%

Example 7: Word Problem Involving Fractions

Question:

Anna eats $\frac{3}{4}$ of a pizza, and her friend Sarah eats $\frac{1}{2}$ of a pizza. How much pizza did they eat in total?

Solution:

- Find a common denominator (in this case, 4).
 $\frac{1}{2}$ is equivalent to $\frac{2}{4}$.
- Now, add the fractions:
 $\frac{3}{4} + \frac{2}{4} = \frac{5}{4}$.
- $\frac{5}{4}$ is an improper fraction, which can be written as $1 \frac{1}{4}$.

Answer: $1 \frac{1}{4}$ pizzas

Practice Questions

Here are some more practice questions to help your child develop their fraction skills:

Practice Question 1: Adding Fractions

What is $\frac{2}{5} + \frac{3}{5}$?

Practice Question 2: Multiplying Fractions

What is $\frac{1}{3} \times \frac{2}{5}$?

Practice Question 3: Dividing Fractions

What is $\frac{4}{7} \div \frac{2}{3}$?

Practice Question 4: Converting Fractions to Decimals

Convert $\frac{7}{20}$ to a decimal.

Practice Question 5: Converting Fractions to Percentages

Convert $\frac{9}{10}$ to a percentage.

Percentages

Percentages can sometimes seem more complex than fractions or decimals, but in reality, they are just a different way of expressing parts of a whole. Once you understand the basic concepts and strategies, percentage questions will become easier to solve. The key is to focus on how the numbers relate to one another and apply simple operations to get the answer.

Below, I'll break down strategies for handling percentage-based questions that may appear on the 11+ GL Maths exam, and explain them in a way that's both easy to understand and easy to apply.

Understanding Percentages

A percentage is just a fraction of 100. The term "percent" literally means "per hundred", so 20% is the same as 20 out of 100 or $\frac{1}{5}$. To solve percentage problems, your child needs to translate percentages into simple operations like multiplication and division.

Basic Strategies for Percentage Questions

1. Converting Percentages to Fractions or Decimals:

- $20\% = \frac{1}{5}$ or 0.20
- $50\% = \frac{1}{2}$ or 0.50
- $25\% = \frac{1}{4}$ or 0.25
- $75\% = \frac{3}{4}$ or 0.75
- $10\% = \frac{1}{10}$ or 0.10

2. Finding a Percentage of a Number:

When you need to find a percentage of a number, multiply the number by the

percentage expressed as a decimal.

Example: To find 20% of 150, multiply 150 by 0.20.

$$150 \times 0.20 = 30$$

3. Finding the Original Price After a Discount:

In problems where a percentage is subtracted (e.g., a discount), first find the percentage of the original price and then subtract it from the total.

Steps:

- Find the percentage of the number.
- Subtract that from the original number.

Example 1: Finding the Percentage of a Number

Question:

What is 20% of £212?

Solution:

- We know that 20% is equivalent to $\frac{1}{5}$.
- To find 20% of 212, divide 212 by 5.
 $212 \div 5 = 42.4$
- Therefore, 20% of £212 is £42.40.

Answer: £42.40

Example 2: Percentage Discount Calculation

Question:

A bed costs £212, and you get a 20% discount. What is the price after the discount?

Solution:

- We know that 20% of £212 is £42.40 (as we calculated earlier).
- To find the price after the discount, subtract £42.40 from the original price of £212.
 $£212 - £42.40 = £169.60$.

Answer: £169.60 (option C)

Types of Percentage Questions on the 11+ GL Exam

1. Percentage of a Number

This is a basic question where you are asked to find a certain percentage of a

given number.

Example: What is 15% of 300?

Solution:

$$300 \times 0.15 = 45$$

2. Percentage Increase or Decrease

Questions can ask for the percentage increase or decrease between two numbers.

Example: A book costs £10, and after a sale, the price becomes £8. What is the percentage decrease?

Solution:

- Find the difference: $£10 - £8 = £2$
- Divide by the original amount: $£2 \div £10 = 0.20$
- Multiply by 100 to get the percentage: $0.20 \times 100 = 20\%$ decrease.

3. Percentage of an Amount After Increase or Decrease

Sometimes, questions will ask you to find the new amount after applying a percentage increase or decrease.

Example: A shop increases the price of an item by 25%. If the original price is £80, what is the new price?

Solution:

- Find 25% of £80: $80 \times 0.25 = £20$
- Add the increase to the original price: $£80 + £20 = £100$.

4. Finding the Original Amount Given the Percentage

In some cases, the question will give you the amount after a percentage increase or decrease and ask you to find the original amount.

Example: After a 20% discount, the price of a jacket is £48. What was the original price?

Solution:

- Let the original price be X .
- We know that 20% of X was taken off, so the price now is 80% of X .
- Set up the equation: $0.80 \times X = £48$
- Solve for X : $X = £48 \div 0.80 = £60$.

Tips for Tackling Percentage Questions

1. Remember the Relationship Between Fractions, Decimals, and Percentages:

- $20\% = 1/5 = 0.20$
- $50\% = 1/2 = 0.50$
- $25\% = 1/4 = 0.25$

2. Look for Key Words:

- "Of" often means multiplication.
- "Discount" means subtract.
- "Increase" means add.
- "Decrease" means subtract.

3. Work Backwards:

If you're given the price after a percentage discount or increase and need to find the original price, divide the final amount by the remaining percentage (after the discount or increase).

4. Use the Same Percentage for Multiple Questions:

Once you know the percentage of a number, try applying it to different situations to speed up the process. For example, if you know that 10% of a number is £10, then 50% will be £50, and 20% will be £20.

5. Double-Check Your Work:

Since percentages often involve converting between fractions, decimals, and whole numbers, it's easy to make small errors. Always double-check your calculations, especially when converting between formats.

Practice Questions

Here are some additional practice questions to help your child master percentages:

1. What is 15% of £200?
2. A shop has a 30% sale. If an item originally costs £120, what is the new price?
3. A school has 500 students. 25% of them are absent on a given day. How many students are absent?
4. The price of a product increases from £40 to £50. What is the percentage increase?
5. After a 10% discount, a jacket costs £72. What was the original price?

Conclusion

With percentages, the process often involves translating the percentage into a more workable form (like a fraction or decimal) and applying it to the problem. Whether it's finding a percentage of a number, calculating a percentage increase or decrease, or working out the original amount, the key to success is practice and confidence.

Special Numbers

This section covers key concepts around special numbers. Understanding these definitions and how to apply them is crucial for success in your child's 11 Plus exam. These include:

Even and Odd Numbers

- Even numbers: These are numbers that can be divided by 2 without leaving a remainder. For example, 2, 4, 6, and 8 are even numbers.
- Odd numbers: These are numbers that cannot be divided by 2 without leaving a remainder. For example, 1, 3, 5, and 7 are odd numbers.

Key Concept: Odd and even numbers alternate. So, after every even number comes an odd number, and vice versa.

Prime Numbers

Prime numbers are numbers greater than 1 that have only two factors: 1 and themselves. Some examples of prime numbers include:

- 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31...

Key Concept: 1 is not a prime number because it only has one factor.

Square Numbers

A square number is a number that is the result of multiplying an integer by itself. For example:

- $4^2 = 4 \times 4 = 16$
- $5^2 = 5 \times 5 = 25$

Key Concept: Square numbers can also be presented with the exponent notation (e.g., 4^2 or 5^2).

Square Roots

The square root of a number is the number that, when multiplied by itself, gives the original number. For example:

- $\sqrt{9} = 3$ because $3 \times 3 = 9$
- $\sqrt{16} = 4$ because $4 \times 4 = 16$

Key Concept: The square root is the opposite operation of squaring a number.

Cube Numbers

A cube number is a number that is the result of multiplying an integer by itself three times. For example:

- $2^3 = 2 \times 2 \times 2 = 8$
- $3^3 = 3 \times 3 \times 3 = 27$

Key Concept: Cube numbers can be presented as 2^3 , 3^3 , and so on.

- To find the cube root, for example: $\sqrt[3]{8} = 2$ because $2 \times 2 \times 2 = 8$.

Triangular Numbers

Triangular numbers are numbers that can be arranged to form a triangle. The first few triangular numbers are:

- 1, 3, 6, 10, 15, 21...

Key Concept: The difference between consecutive triangular numbers increases by one each time.

For example, between 3 and 6 there's a difference of 3, and between 6 and 10 there's a difference of 4.

Indices

Indices (also called exponents) show how many times a number is multiplied by itself. For example:

- $6^5 = 6 \times 6 \times 6 \times 6 \times 6 = 7776$
- Here, the index is 5, meaning we multiply 6 by itself 5 times.

Key Concept: Indices help simplify large calculations.

Roman Numerals

Roman numerals are a system of numbers used in ancient Rome. Some key Roman numerals are:

- I = 1
- II = 2
- III = 3
- IV = 4
- V = 5
- VI = 6
- VII = 7

- VIII = 8
- IX = 9
- X = 10

Key Concept: Remember that 4 is not written as IIII, but IV, and 9 is written as IX (not VIIII).

Negative Numbers

Negative numbers are numbers that are less than zero.

For example:

- -1, -2, -3, -4, -5...

Key Concept: When multiplying or dividing two negative numbers, the result is positive (e.g., $-2 \times -3 = 6$). But when a negative number is multiplied or divided by a positive number, the result is negative (e.g., $-2 \times 3 = -6$).

Consecutive Numbers

Consecutive numbers are numbers that follow one another in order. For example:

- 1, 2, 3, 4, 5 or
- 56, 57, 58, 59...

Key Concept: Questions involving consecutive numbers might ask you to find a set of numbers that add up to a specific total. For example:

- Find 5 consecutive numbers that add up to 45.

Solution:

1. Divide 45 by 5:
 $45 \div 5 = 9$
2. Check numbers around 9:
 $7 + 8 + 9 + 10 + 11 = 45$
So, the correct numbers are 7, 8, 9, 10, and 11.

Example Question: Prime Numbers

- The first five prime numbers are 2, 3, 5, 7, and 11.
- Their sum is:
 $2 + 3 + 5 + 7 + 11 = 28$.
- We now need to add the next two prime numbers:
13 and 17.
 $28 + 13 + 17 = 58$.

Therefore, the correct answer is 58, which corresponds to option A.

Key Takeaways:

- Special numbers, such as prime numbers, square numbers, cube numbers, and triangular numbers, play a crucial role in many 11+ questions.
- Understanding indices, negative numbers, and Roman numerals are vital for solving a wide range of problems in both the maths and reasoning sections.
- Knowing the rules for consecutive numbers and how to apply ratios, proportions, and scale can make a significant difference in solving tricky problems.

Angles and Lines

Understanding angles, triangles, and lines is essential for solving many geometry-based problems in the 11 Plus exam. These types of questions often involve recognizing different angles formed by polygons, calculating interior and exterior angles, and applying geometric properties to find unknown angles.

Key Concepts in Angles and Lines:

1. Types of Angles:

- Acute angles: Angles less than 90° .
- Right angles: Angles exactly equal to 90° .
- Obtuse angles: Angles greater than 90° but less than 180° .
- Straight angles: Angles exactly equal to 180° .
- Reflex angles: Angles greater than 180° but less than 360° .

2. Angle Sum Properties:

- The sum of the interior angles of any triangle is always 180° .
- The sum of the interior angles of a quadrilateral is always 360° .

2D Shapes

Questions involving 2D shapes typically require you to identify different shapes based on their properties or descriptions. Understanding the characteristics of various 2D shapes is crucial for recognizing them in different forms, whether they are drawn or described verbally. Additionally, knowledge of these shapes will often assist with related questions about areas, perimeters, and angles within these shapes.

Key 2D Shapes to Know:

1. Triangles:

- Equilateral Triangle: All three sides and angles are equal (each angle = 60°).
- Isosceles Triangle: Two sides and two angles are equal.
- Scalene Triangle: All sides and angles are different.

2. Quadrilaterals:

- Square: Four equal sides and four right angles.
- Rectangle: Opposite sides are equal, and all angles are 90° .
- Rhombus: All sides are equal, but angles are not 90° . Opposite angles are equal, and opposite sides are parallel.
- Parallelogram: Opposite sides are equal and parallel, and opposite angles are equal.
- Trapezium (UK Definition): A quadrilateral with one pair of parallel sides.
- Kite: Two pairs of adjacent sides are equal. The diagonals intersect at right angles, and one diagonal bisects the other.

3. Circles:

- Circle: A shape with all points equidistant from the center.

4. Other Polygons:

- Pentagon: A polygon with five sides.
- Hexagon: A polygon with six sides.
- Octagon: A polygon with eight sides.

Example Problem: Identifying Shapes

Problem: Ranjith draws a shape. It has one pair of parallel sides and two pairs of adjacent equal sides. What type of shape is this?

Solution:

Let's break down the description:

- One pair of parallel sides: This means the shape must be a trapezium or a parallelogram. Other shapes like squares, rectangles, rhombuses, and kites have at least two pairs of parallel sides.
- Two pairs of adjacent equal sides: This is a key characteristic of a kite or rhombus.

However, if we carefully consider the description, we can rule out the kite because a kite does not have parallel sides. We are left with two possible shapes: trapezium and parallelogram.

But because the shape only has one pair of parallel sides, it cannot be a parallelogram (which always has two pairs of parallel sides). Therefore, the shape described must be a trapezium.

So, the answer is C, Trapezium.

Additional Key Concepts and Properties:

1. Properties of a Square:

- Four equal sides and four right angles.
- The diagonals bisect each other at right angles and are of equal length.
- A square is also a special case of a rectangle and a rhombus.

2. Properties of a Rectangle:

- Opposite sides are equal and parallel, with all angles being 90° .
- The diagonals are equal in length but do not necessarily bisect each other at right angles.

3. Properties of a Rhombus:

- All four sides are equal, but the angles are not 90° .
- Opposite angles are equal, and opposite sides are parallel.
- The diagonals bisect each other at right angles.

4. Properties of a Parallelogram:

- Opposite sides are parallel and equal in length.
- Opposite angles are equal.
- The diagonals bisect each other but are not necessarily of equal length.

5. Properties of a Kite:

- Two pairs of adjacent equal sides.
- The diagonals intersect at right angles, and one of the diagonals bisects the other.
- One diagonal acts as a line of symmetry, splitting the kite into two equal parts.

Practice Question 1:

Problem:

Which of the following shapes has two pairs of parallel sides, opposite angles equal, and diagonals that bisect each other?

Options: A) Square
B) Rectangle

- C) Rhombus
- D) Parallelogram

Solution: The description matches the properties of a parallelogram, as all of these features are characteristic of it. A square, rectangle, and rhombus also have two pairs of parallel sides, but the diagonals bisect each other specifically in a parallelogram.

So, the correct answer is D) Parallelogram.

Practice Question 2:

Problem:

Which shape has one pair of parallel sides and two pairs of adjacent equal sides?

- Options:
- A) Trapezium
 - B) Parallelogram
 - C) Kite
 - D) Square

Solution: The key description is one pair of parallel sides and two pairs of adjacent equal sides, which clearly describes a kite.

Therefore, the correct answer is C) Kite.

Practice Question 3:

Problem:

Which shape has four equal sides, opposite angles equal, and the diagonals bisect each other at right angles?

- Options:
- A) Rectangle
 - B) Rhombus
 - C) Square
 - D) Parallelogram

Solution: This description fits a square. A square has four equal sides, opposite angles equal, and its diagonals bisect each other at right angles.

So, the correct answer is C) Square.

Strategies for Solving 2D Shapes Questions:

1. Know the Key Properties:

- Memorize the defining properties of each shape. For example, squares have all equal sides and right angles, while rhombuses have equal sides but angles that are not necessarily right angles.

2. Eliminate Incorrect Options:

- In multiple-choice questions, often the easiest way to find the correct shape is by eliminating the shapes that don't match the description. For example, if a shape has no parallel sides, it can't be a parallelogram or rectangle.

3. Draw the Shape:

- When unsure about a question, try sketching the shape based on the description. This can help visualize the relationships between the sides and angles.

4. Use the Symmetry Properties:

- Shapes like squares, rectangles, and kites have symmetry. Understanding how the shape behaves symmetrically can often lead to the correct answer.

3D Shapes

Understanding 3D shapes is an essential skill for the 11 Plus exam. These types of questions often require you to identify and work with different shapes such as cubes, cylinders, cones, spheres, pyramids, and prisms. The ability to visualize these shapes mentally, especially when they are described in three dimensions, is crucial. While it's easy to see how 2D shapes fit together, 3D shapes may require a bit more imagination, especially when we're working with problems where the shapes aren't drawn accurately on paper or are partially hidden.

Key 3D Shapes to Know:

1. Cube:

- All sides are square.
- All angles are right angles.
- 6 faces, 12 edges, 8 vertices.

2. Cuboid (Rectangular Prism):

- Opposite faces are rectangular.
- 6 faces, 12 edges, 8 vertices.

3. Sphere:

- All points on the surface are equidistant from the center.
- No edges or vertices.
- 4. Cylinder:
 - Two circular faces, one at the top and one at the bottom.
 - The curved surface connects these two faces.
- 5. Cone:
 - A circular base with a pointed top.
 - One curved surface and one circular face.
- 6. Pyramid:
 - A polygonal base with triangular faces that meet at a point (apex).
 - The number of faces equals the number of sides on the base.
- 7. Prism:
 - A polyhedron with two congruent bases and parallelogram faces connecting the corresponding sides.
 - The shape of the base determines the name (e.g., triangular prism, rectangular prism).

Example Problem: Rubik's Cube and Paint

Problem: We are shown a 3x3x3 Rubik's Cube, which is made up of 27 smaller cubes. The cube is dropped into paint, and we are asked to determine how many of the smaller cubes will only have one face painted.

Solution:

Let's break it down:

- A 3x3x3 Rubik's Cube consists of 27 smaller cubes in total (3 cubes along each edge, 3 edges per face, and 3 faces on the cube: $3 \times 3 \times 3 = 27$).
- We are interested in finding the smaller cubes that have only one face painted.
- To solve this, we need to visualize how the Rubik's Cube is structured:
 - The Rubik's Cube has 6 faces.
 - The cubes at the corners of the Rubik's Cube have three faces exposed (and painted).
 - The cubes along the edges (excluding the corners) have two faces exposed (and painted).

- The cubes that are in the center of each face have only one face exposed.
- Since we have a 3x3 grid on each face, the center cube on each face is the one with only one face exposed.
- There are 6 faces on the Rubik's Cube, so there will be 6 cubes, one from each face, that have only one face painted.
- Therefore, the answer is C) 6.

Further Example: Cube and Paint

Problem: A 4x4x4 cube is made up of 64 smaller cubes. The cube is painted on the outside and then broken into its smaller cubes. How many of the smaller cubes have no paint on them at all?

Solution:

- A 4x4x4 cube consists of 64 smaller cubes (4 cubes along each edge: $4 \times 4 \times 4 = 64$).
- The smaller cubes with no paint on them must be the ones that are entirely inside the cube and not touching the surface.
- If we remove one layer of cubes from the outside of the 4x4x4 cube (which is painted), we are left with a smaller 3x3x3 cube inside. This smaller cube consists of 27 cubes ($3 \times 3 \times 3 = 27$).
- These 27 cubes are the only ones without any paint on them, as they are all inside the larger cube and are not exposed to the surface.
- Therefore, the answer is 27.

Key Tips for 3D Shape Questions:

1. Visualize the Shape:
 - It's important to mentally visualize how the 3D shapes are constructed. For example, in problems involving a cube or a Rubik's Cube, imagine cutting the cube into smaller cubes and considering which ones are exposed and which are not.
2. Understand Exposed Faces:

- Focus on how many faces of the 3D shape are exposed (such as the corners, edges, or the center of each face). This is especially useful when dealing with problems that involve paint or surface areas.
3. Look for Symmetry:
- Many 3D shapes have symmetrical properties that can help you find the solution quickly. For instance, a cube has identical faces and will behave symmetrically in most cases

Area and Perimeter (Compound Shapes)

In the 11 Plus exam, you may encounter questions involving compound shapes. These shapes are made up of two or more simple shapes (such as squares, rectangles, or circles) combined together. To solve these problems, you need to calculate the area or perimeter of each individual shape and then combine the results, either by adding or subtracting depending on how the shapes are arranged.

Key Formulas to Remember

- Area of a Rectangle:
 $\text{Area} = \text{length} \times \text{width}$
- Perimeter of a Rectangle:
 $\text{Perimeter} = 2 \times (\text{length} + \text{width})$
- Area of a Square:
 $\text{Area} = \text{side} \times \text{side} = \text{side}^2$
- Perimeter of a Square:
 $\text{Perimeter} = 4 \times \text{side}$

Example Problem 1: Area of a Compound Shape

Problem: A compound shape consists of a rectangle with a length of 6m and a width of 4m, and a square with a side length of 4m attached to one of the sides of the rectangle. What is the total area?

Solution:

1. Step 1: Calculate the area of the rectangle
The area of the rectangle is:
 $\text{Area} = \text{length} \times \text{width} = 6\text{m} \times 4\text{m} = 24\text{m}^2$

2. Step 2: Calculate the area of the square

The area of the square is:

$$\text{Area} = \text{side} \times \text{side} = 4\text{m} \times 4\text{m} = 16\text{m}^2$$

3. Step 3: Total area of the compound shape

To find the total area, we add the areas of the rectangle and the square:

$$\text{Total Area} = 24\text{m}^2 + 16\text{m}^2 = 40\text{m}^2$$

So, the total area of the compound shape is 40m^2 .

Example Problem 2: Perimeter of a Compound Shape

Problem: A compound shape is made up of a rectangle with a length of 8m and a width of 5m, and a square with a side length of 5m attached to one side of the rectangle. What is the perimeter of the compound shape?

Solution:

1. Step 1: Calculate the perimeter of the rectangle

The perimeter of the rectangle is:

$$\text{Perimeter} = 2 \times (\text{length} + \text{width}) = 2 \times (8\text{m} + 5\text{m}) = 2 \times 13\text{m} = 26\text{m}$$

2. Step 2: Calculate the perimeter of the square

The perimeter of the square is:

$$\text{Perimeter} = 4 \times \text{side} = 4 \times 5\text{m} = 20\text{m}$$

3. Step 3: Adjust for the side where the square is attached

Since one side of the square is attached to the rectangle, we need to subtract that side from the total perimeter. The side of the square is 5m, so:

$$\text{Total Perimeter} = 26\text{m} + 20\text{m} - 5\text{m} = 41\text{m}$$

Thus, the perimeter of the compound shape is 41m.

Practice Question 1: Area of a Compound Shape

Problem:

A compound shape consists of a rectangle with a length of 10m and a width of 4m, and a square with a side length of 4m attached to one of the longer sides of the rectangle. What is the total area?

Solution:

1. Step 1: Area of the rectangle

$$\text{Area} = 10\text{m} \times 4\text{m} = 40\text{m}^2$$

2. Step 2: Area of the square
 $\text{Area} = 4\text{m} \times 4\text{m} = 16\text{m}^2$
3. Step 3: Total area
 $\text{Total Area} = 40\text{m}^2 + 16\text{m}^2 = 56\text{m}^2$

Practice Question 2: Perimeter of a Compound Shape

Problem:

A compound shape is made up of a rectangle with a length of 7m and a width of 3m, and a square with a side length of 3m attached to one of the shorter sides of the rectangle. What is the perimeter of the compound shape?

Solution:

1. Step 1: Perimeter of the rectangle
 $\text{Perimeter} = 2 \times (\text{length} + \text{width}) = 2 \times (7\text{m} + 3\text{m}) = 2 \times 10\text{m} = 20\text{m}$
2. Step 2: Perimeter of the square
 $\text{Perimeter} = 4 \times 3\text{m} = 12\text{m}$
3. Step 3: Adjust for the side where the square is attached
We need to subtract the side of the square that is attached to the rectangle.
The side of the square is 3m, so:
 $\text{Total Perimeter} = 20\text{m} + 12\text{m} - 3\text{m} = 29\text{m}$

Key Tips for Compound Shapes

1. Break it down into simpler shapes:
Divide the compound shape into rectangles, squares, or other simple shapes.
This will help you work out the area or perimeter more easily.
2. Calculate area first, then perimeter:
For compound shapes, it's often easier to start with the area, and then move on to the perimeter. For perimeter, be sure to adjust for any sides that are shared between two shapes.
3. Watch out for overlapping shapes:
When shapes overlap, you might need to subtract certain sides or parts to avoid double counting.

Measurement

Measurement questions in the 11 Plus exam test two key areas:

1. Calculations involving length, mass, volume, and capacity.
2. Understanding different units of measurement, including both metric and imperial systems.

To solve these problems, it's important to have a good grasp of the units used in both systems and be comfortable converting between them when necessary.

Additionally, many of these questions involve basic arithmetic, such as multiplication and division.

Example Problem: Cost Per Mile Calculation

Problem:

A 110-mile journey uses 22 litres of fuel. The cost of fuel is £1.15 per litre. What is the cost per mile of this journey?

Solution:

1. Step 1: Calculate the total fuel cost
To find out the total cost of the fuel for the journey, multiply the cost per litre by the number of litres used:
 $£1.15 \times 22 = £25.3$.
2. Step 2: Calculate the cost per mile
Now, to find the cost per mile, divide the total fuel cost by the distance travelled (in miles):
 $£25.3 \div 110 = £0.23$ (rounded to two decimal places).

So, the cost per mile is £0.23, which corresponds to option E.

Key Concepts in Measurement Questions

1. Metric Units:

- Length: millimetres (mm), centimetres (cm), metres (m), kilometres (km)
- Mass: milligrams (mg), grams (g), kilograms (kg), tonnes (t)
- Volume/Capacity: millilitres (ml), litres (l), cubic centimetres (cm³), cubic metres (m³)

2. Imperial Units:

- Length: inches (in), feet (ft), yards (yd), miles (mi)
- Mass: ounces (oz), pounds (lbs), stones (st)
- Volume/Capacity: fluid ounces (fl oz), pints (pt), quarts (qt), gallons (gal)

3. Conversions between Metric and Imperial Units

It's important to know how to convert between these systems. For example:

- 1 mile = 1.609 kilometres
- 1 kilogram = 2.20462 pounds
- 1 litre = 0.21997 gallons (UK), 0.26417 gallons (US)

4. Units of Speed (for distance-related problems):

When dealing with journeys, speed is often calculated in miles per hour (mph) or kilometres per hour (km/h).

Practice Question 1: Volume Calculation

Problem:

A container holds 5 litres of water. If a person uses 200 millilitres (ml) of water every day, how many days will the water last?

Solution:

1. Step 1: Convert 5 litres into millilitres
Since 1 litre = 1000 millilitres,
5 litres = $5 \times 1000 = 5000$ millilitres.
2. Step 2: Calculate the number of days
Now, divide the total volume by the daily usage:
 $5000 \div 200 = 25$ days.

So, the water will last 25 days.

Practice Question 2: Mass Calculation

Problem:

A bag of flour weighs 2.5 kilograms. How many grams is this?

Solution:

1. Step 1: Convert kilograms to grams
Since 1 kilogram = 1000 grams,
2.5 kilograms = $2.5 \times 1000 = 2500$ grams.

So, the weight of the flour is 2500 grams.

Tips for Measurement Questions:

1. Know your conversions:

Be sure to memorise common conversions between metric and imperial units, such as:

- 1 inch = 2.54 cm
- 1 mile = 1.609 km
- 1 kilogram = 2.2046 pounds

2. Work methodically:

Measurement problems often involve multiple steps. Break the problem into manageable parts, and don't forget to check units at each stage.

3. Pay attention to rounding:

For some questions, especially those involving currency or measurements, round your final answer to the required number of decimal places (e.g., 2 decimal places for money).

4. Understanding word problems:

Read each problem carefully to identify the quantities involved and the units of measurement. The most important part of solving measurement questions is understanding what you're being asked to find and what information is given.

Marking & Interpreting Scales

In these types of questions, you are asked to interpret a number line or scale, often without being explicitly told what the scale is. You'll need to figure out the intervals between the numbers based on the information given, such as the values at either end of the scale and how many divisions (or sections) there are in between.

Example Problem: Identifying Time on a Number Line

Problem:

A number line is drawn between 5:00pm and 6:00pm, with an arrow pointing to a certain time. We are asked to identify the time corresponding to the arrow.

Solution:

1. Step 1: Work out the scale of the number line

The number line is between 5:00pm and 6:00pm, and it's divided into 10 equal parts.

This means there are 10 subdivisions between the two times.

2. Step 2: Calculate the interval for each subdivision

Since there are 60 minutes between 5:00pm and 6:00pm, divide 60 by 10 (the number of subdivisions):

$60 \div 10 = 6$ minutes per subdivision.

3. Step 3: Identify the time corresponding to the arrow
The arrow is pointing 3 subdivisions before 6:00pm.
Since each subdivision represents 6 minutes, we calculate:
 $3 \times 6 = 18$ minutes before 6:00pm.
4. Step 4: Calculate the time
To find the time 18 minutes before 6:00pm, subtract 18 minutes from 6:00pm:
 $6:00\text{pm} - 18 \text{ minutes} = 5:42\text{pm}$.

Therefore, the time marked by the arrow is 5:42pm, which corresponds to option D.

Key Concepts for Marking & Interpreting Scales:

1. Identify the Range:
Look at the numbers on either end of the scale. These represent the start and end points of the scale. For example, a number line might range from 5:00pm to 6:00pm, from 0 to 100, or from 1 to 10.
2. Count the Divisions:
Count how many subdivisions are between the two end points. This could be 10 sections, 5 sections, or any other number.
3. Determine the Scale:
To find out what each division represents, divide the total range by the number of subdivisions. For example:
 - If the scale goes from 0 to 100 and there are 10 subdivisions, each subdivision represents 10 ($100 \div 10 = 10$).
 - If the scale goes from 5:00pm to 6:00pm and there are 10 subdivisions, each subdivision represents 6 minutes ($60 \div 10 = 6$).
4. Interpret the Arrow:
Once you know the scale, count the number of subdivisions from the start or end point to where the arrow is pointing. Multiply the number of subdivisions by the value each subdivision represents to find the difference from the start or end point.
5. Final Calculation:
After determining the value corresponding to the arrow, you may need to add or subtract this value from the start or end point to find the final answer.

Practice Question 1: Identifying the Number

Problem:

A number line goes from 20 to 40, with 4 equal subdivisions. An arrow is pointing to one of the subdivisions. What number does the arrow point to?

Solution:

1. Step 1: Work out the scale of the number line
The number line goes from 20 to 40, so the range is $40 - 20 = 20$.
2. Step 2: Find the value for each subdivision
Since there are 4 subdivisions, divide the total range (20) by 4:
 $20 \div 4 = 5$.
3. Step 3: Identify the number for each subdivision
Starting from 20:
 - The first subdivision will be $20 + 5 = 25$.
 - The second subdivision will be $25 + 5 = 30$.
 - The third subdivision will be $30 + 5 = 35$.
 - The fourth subdivision will be $35 + 5 = 40$.

If the arrow is pointing to the second subdivision, the answer is 30.

Practice Question 2: Interpreting the Scale

Problem:

A number line goes from 7:00am to 8:00am, divided into 12 equal parts. An arrow is pointing to the 5th subdivision. What time does the arrow indicate?

Solution:

1. Step 1: Work out the scale of the number line
The range is from 7:00am to 8:00am, so there are 60 minutes between 7:00am and 8:00am.
2. Step 2: Find the value for each subdivision
Divide the total range (60 minutes) by the 12 subdivisions:
 $60 \div 12 = 5$ minutes per subdivision.
3. Step 3: Identify the time for the 5th subdivision
Starting from 7:00am:
 - The first subdivision is 7:05am.
 - The second subdivision is 7:10am.
 - The third subdivision is 7:15am.
 - The fourth subdivision is 7:20am.

- The fifth subdivision is 7:25am.

Therefore, the time indicated by the arrow is 7:25am.

Tips for Marking & Interpreting Scales:

1. Always identify the total range: Start by looking at the end points of the scale. This will give you the range over which the scale is spread.
2. Work out the value of each subdivision: Divide the total range by the number of divisions to find out what each section represents.
3. Count the number of subdivisions: Carefully count how many subdivisions the arrow is pointing to, then multiply this by the value of each subdivision.
4. Use simple arithmetic: Once you've determined the scale, the rest is just basic addition or subtraction, depending on whether the arrow is before or after the start point.

Coordinates

Coordinates-based questions test your child's ability to locate points on a grid, as well as to perform transformations such as rotations, translations, and reflections of shapes. These are key skills in geometry and often appear in the 11+ exam.

Key Concepts to Remember:

1. Coordinates System:

- The grid is split into four quadrants with the origin (0, 0) at the centre.
- X-axis runs horizontally (left to right).
- Y-axis runs vertically (up and down).
- Coordinates are written as (X, Y).
 - X represents the position along the X-axis (left or right).
 - Y represents the position along the Y-axis (up or down).

2. For example, the point (4, 3) means 4 units to the right on the X-axis, and 3 units up on the Y-axis.

3. Rotation:

- 90° rotation: Turn the shape 1 quarter of a full turn, which swaps the X and Y values and changes the sign of the new X.
- 180° rotation: Turn the shape halfway around, which changes the signs of both the X and Y values.
- 270° rotation: Turn the shape 3 quarters of a full turn, which swaps the X and Y values and changes the sign of the new Y.

4. Translation:

- Translation means moving a shape a certain number of units up, down, left, or right without rotating it.
- For example, translating a point (3, 4) 2 units up and 4 units right will result in the new coordinates (7, 6).

5. Reflection:

- Reflection involves flipping the shape over a certain axis. Common reflections are across the X-axis or Y-axis.
- Reflection over the X-axis: The X values stay the same, but the Y values change sign.
- Reflection over the Y-axis: The Y values stay the same, but the X values change sign.

Example Problem: Identifying Coordinates

Problem:

In the example, we need to identify the coordinates of a water fountain located in a quadrant. We are given multiple options for the possible coordinates, and we need to select the correct one.

- The X-axis value represents how far across the point is from the vertical line.
- The Y-axis value represents how far up the point is from the horizontal line.

Solution:

- The coordinates of the water fountain are represented by (4, 3).
- $X = 4$ means the point is 4 units to the right on the X-axis.
- $Y = 3$ means the point is 3 units up on the Y-axis.

Thus, the correct coordinates for the water fountain are (4, 3). According to the options, this is option C.

Strategies for Solving Coordinates Questions:

1. Understand the Grid:

Always recall that the X-coordinate comes first, followed by the Y-coordinate. The X-coordinate tells you how far the point is along the horizontal axis, and the Y-coordinate tells you how far the point is along the vertical axis.

2. Practice Identifying Points:

In many 11+ questions, you'll be asked to identify the coordinates of a point on a grid. Practice by mentally counting the squares along the X and Y axes. Check that you're counting from the origin (0, 0) and moving right for positive X, up for positive Y.

3. Rotations:

To rotate a shape, remember the rules:

- 90° rotation swaps the X and Y values, changing the sign of the new X.
- 180° rotation changes the signs of both the X and Y values.
- 270° rotation swaps the X and Y values, changing the sign of the new Y.

4. It may help to visualize this or use graph paper to practice.

5. Translations:

For translation, you simply move the shape by a given number of units along the X and Y axes.

- Moving to the right means adding to the X value.
- Moving to the left means subtracting from the X value.
- Moving up means adding to the Y value.
- Moving down means subtracting from the Y value.

6. Reflections:

To reflect a point across the X-axis or Y-axis:

- Reflection across the X-axis: Keep the X value the same, but change the sign of the Y value.
- Reflection across the Y-axis: Keep the Y value the same, but change the sign of the X value.

Practice Questions:

Question 1: Identify the coordinates of a point.

A point is located 3 units to the right of the origin and 4 units up. What are the coordinates?

Solution:

The coordinates are (3, 4).

- 3 units to the right means the X value is 3.
- 4 units up means the Y value is 4.

Question 2: Apply a 90° rotation.

A point is located at (2, 5). What are the new coordinates after a 90° rotation?

Solution:

To rotate (2, 5) by 90° , swap the X and Y values and change the sign of the new X.
New coordinates: (-5, 2).

Question 3: Translate a point.

A point is located at (4, 3). Translate it 2 units to the left and 3 units down. What are the new coordinates?

Solution:

- Moving 2 units left means subtract 2 from the X value: $4 - 2 = 2$.
- Moving 3 units down means subtract 3 from the Y value: $3 - 3 = 0$.
The new coordinates are (2, 0).

Question 4: Reflection across the Y-axis.

A point is located at (7, -4). What are the coordinates after reflecting it across the Y-axis?

Solution:

- Reflecting across the Y-axis changes the sign of the X value, while the Y value stays the same.
The new coordinates are (-7, -4).

Summary:

Coordinate questions test your ability to read and interpret points on a grid, as well as apply transformations such as rotation, translation, and reflection. Key strategies include:

- Always remember the order of coordinates: (X, Y).
- Use rules for rotation, translation, and reflection to accurately transform points.
- Practice using graph paper or visualizing transformations to improve speed and accuracy.

Mean, Mode, Median & Range

Questions about mean, mode, median, and range often appear in the 11+ exam and involve analyzing a set of numbers or data points. Here's a quick breakdown of what each term means:

Definitions:

1. Mean:

- The mean is the average of a set of numbers.

- To calculate the mean, add up all the numbers in the set, then divide that total by how many numbers there are.
- 2. Formula:
Mean
= Sum of all values/ Number of values
- 3. Mode:
 - The mode is the number that appears the most frequently in a data set.
 - There may be no mode, one mode, or multiple modes if different numbers appear the same maximum number of times.
- 4. Median:
 - The median is the middle number when the data set is ordered from lowest to highest.
 - If there is an even number of values, the median is the average of the two middle numbers.
- 5. Range:
 - The range is the difference between the largest and smallest numbers in a data set.
- 6. Formula:
Range=Maximum value–Minimum value

Example Problem:

Question:

The mean height of a group of children is 140 cm. This mean height is 7 times the range of the heights. The height of the tallest child is 164 cm. What is the height of the shortest child?

Solution:

1. Step 1: We know that the mean height is 140 cm, and the mean is 7 times the range.
 - To find the range, divide the mean by 7
2. Step 2: The range is the difference between the tallest and shortest child, so:
Range=Tallest height–Shortest height

3. We are told that the tallest height is 164 cm, and the range is 20 cm. So, we can write the equation:
 $164 - \text{Shortest height}$

4. Step 3: Solve for the Shortest height:

$$\text{Shortest height} = 164 - 20 = 144 \text{ cm}$$

Therefore, the height of the shortest child is 144 cm. The correct answer is option A.

Practice Questions:

Question 1:

The scores of five students in a test are: 58, 72, 80, 85, and 90. What is the mean score?

Solution:

1. Add all the scores: $58 + 72 + 80 + 85 + 90 = 385$
2. Divide by the number of students (5)

The mean score is 77.

Question 2:

The ages of 6 children are: 6, 8, 12, 12, 15, and 18. What is the mode?

Solution:

The number 12 appears twice, while the other numbers only appear once. Thus, the mode is 12.

Question 3:

The numbers in a set are: 10, 15, 20, 25, 30. What is the median?

Solution:

1. Order the numbers from lowest to highest: 10, 15, 20, 25, 30.
2. The middle number is 20.
Thus, the median is 20.

Question 4:

The data set is: 14, 18, 22, 28, 34. What is the range?

Solution:

1. The largest value is 34, and the smallest value is 14.
2. The range is:
 $\text{Range} = 34 - 14 = 20$

The range is 20.

Summary:

- Mean is the average, found by adding all numbers together and dividing by how many there are.
- Mode is the most frequent number in the set.
- Median is the middle value when the numbers are arranged in order.
- Range is the difference between the highest and lowest values.

Probability (in Percentages)

When dealing with probability in percentages, the principle remains the same as for fractions, but instead of expressing probability as a fraction or decimal, we convert the result into a percentage. Probability in percentages tells us how likely an event is to happen in percentage form, ranging from 0% (impossible) to 100% (certain).

Key Concept for Probability in Percentages:

1. Formula for Probability:
$$\text{Probability} = \frac{\text{Number of favorable outcomes}}{\text{Total number of possible outcomes}}$$

$\text{Probability} = \frac{\text{Number of favorable outcomes}}{\text{Total number of possible outcomes}}$
To express this as a percentage, multiply the result by 100.

2. Steps for finding probability as a percentage:
 - Step 1: Find the number of favorable outcomes.
 - Step 2: Find the total number of possible outcomes.
 - Step 3: Divide the favorable outcomes by the total number of outcomes.

- Step 4: Multiply by 100 to convert the fraction into a percentage.

Example Problem:

Question:

A hexagonal spinner has six sections, numbered 1 to 6. What is the probability that the spinner lands on an odd number?

Solution:

1. Total number of possible outcomes: 6 (since there are 6 sections on the spinner: 1, 2, 3, 4, 5, 6).
2. Favorable outcomes (odd numbers): 1, 3, and 5. There are 3 favorable outcomes.
3. The probability of landing on an odd number is:
 0.5
4. Convert to a percentage:
 $0.5 \times 100 = 50\%$

So, the probability of landing on an odd number is 50%. The correct answer is option B.

Practice Questions with Percentages:

Question 1:

A die is rolled. What is the probability that it shows a number greater than 4?

Solution:

1. Total number of possible outcomes: 6 (since a die has 6 faces: 1, 2, 3, 4, 5, 6).
2. Favorable outcomes (numbers greater than 4): 5 and 6. There are 2 favorable outcomes.
3. The probability is:
 $2/6 = 0.3333$
4. Convert to a percentage:
 $0.3333 \times 100 = 33.33\%$

So, the probability of rolling a number greater than 4 is 33.33%.

Question 2:

A bag contains 5 red balls, 3 blue balls, and 2 green balls. What is the probability of randomly selecting a red ball?

Solution:

1. Total number of balls:
 $5+3+2=10$
2. Favorable outcomes (red balls): 5.
3. The probability is:
 $\frac{5}{10}=0.5$
4. Convert to a percentage:
 $0.5 \times 100 = 50\%$

So, the probability of selecting a red ball is 50%.

Question 3:

A spinner has 8 equal sections: 4 of them are yellow, 2 are blue, and 2 are red. What is the probability that the spinner lands on a blue section?

Solution:

1. Total number of sections: 8.
2. Favorable outcomes (blue sections): 2.
3. The probability is:
 $\frac{2}{8}=0.25$
4. Convert to a percentage:
 $0.25 \times 100 = 25\%$

So, the probability of landing on a blue section is 25%.

Summary:

- Probability as a Percentage: Multiply the fraction (calculated as favorable outcomes divided by total outcomes) by 100.

- Range: Probability can range from 0% (impossible event) to 100% (certain event).
- The probability is usually presented as a percentage for clarity and ease of understanding, especially in everyday contexts like games, spinners, or drawing cards.

Time: 24-Hour Clock and Timetables

1. Breaking Down Time Durations

Strategy: Add or Subtract Time Using Hours and Minutes

- How: When asked to calculate a time duration, break it down into hours and minutes. For example, if something starts at 2:30pm and lasts for 75 minutes, first add the hours (if any) and then add the minutes.
- Example:
Question: "If a meeting starts at 2:30pm and lasts for 75 minutes, what time will it finish?"
 - Break the 75 minutes into 1 hour and 15 minutes.
 - Add 1 hour to 2:30pm = 3:30pm.
 - Add the remaining 15 minutes = 3:45pm.
 - Answer: 3:45pm.

2. Converting Between 12-hour and 24-hour Time

Strategy: Convert AM/PM to 24-Hour Time Format

- How: To convert from 12-hour format to 24-hour format, add 12 to any PM time (except 12:00 PM, which stays the same). For AM times, they stay as they are, except for 12:00 AM which becomes 00:00 (midnight).
- Example:
Question: "What is 3:45pm in 24-hour time?"
 - Since 3:45pm is after noon, add 12 to the hour:
3:45pm → 15:45 in 24-hour time.
 - Answer: 15:45.

3. Calculating Time Across Different Time Zones

Strategy: Adjust for Time Zone Differences

- How: When converting times across time zones, remember the time difference. If you're told one location is 5 hours ahead or behind another, simply add or subtract the difference from the given time.
- Example:
Question: "Abby is in London and Karl is in New York. If Abby calls Karl at

2:30pm London time, what time will it be in New York, where the time zone is 5 hours behind?"

- Abby's call is at 2:30pm in London.
- Subtract 5 hours from 2:30pm:
 $2:30\text{pm} - 5 \text{ hours} = 9:30\text{am}.$
- Answer: 9:30am in New York.

4. Using Timetables Effectively

Strategy: Follow the Sequence on Timetables

- How: When using a timetable, look at the start time of one event, then check the duration between events. If you need to calculate the next event's time, use simple addition or subtraction.
- Example:
Question: "A train departs at 3:15pm and takes 2 hours 45 minutes to reach its destination. What time does it arrive?"
 - Add 2 hours to 3:15pm = 5:15pm.
 - Add the remaining 45 minutes to 5:15pm = 6:00pm.
 - Answer: 6:00pm.

5. Calculating Time Differences (Duration)

Strategy: Subtract Start Time from End Time

- How: To calculate the duration between two times, subtract the start time from the end time. If crossing over the hour, adjust accordingly (e.g., 5:50 to 6:20 = 30 minutes).
- Example:
Question: "The meeting started at 2:30pm and finished at 4:10pm. How long did the meeting last?"
 - From 2:30pm to 4:00pm is 1 hour and 30 minutes.
 - From 4:00pm to 4:10pm is an additional 10 minutes.
 - Answer: 1 hour 40 minutes.

6. Understanding 'AM' and 'PM' Clearly

Strategy: Remember the Difference Between AM and PM

- How: Always distinguish between AM (morning) and PM (afternoon/evening) when calculating times. Be careful with midnight (12:00 AM) and noon (12:00 PM) since they are special cases in time calculations.
- Example:
Question: "What is 12:00 AM in 24-hour format?"

- Answer: 00:00.

7. Double-Check Time Zones

Strategy: Add or Subtract Correctly for Different Zones

- How: Always read carefully when dealing with time zones. If you're given a time zone difference (e.g., +5 hours or -3 hours), make sure to adjust correctly in relation to the starting time.
- Example:
Question: "A flight departs from London at 9:00am and arrives in Sydney, which is 11 hours ahead. What time will it be in Sydney when the flight arrives?"
 - Add 11 hours to 9:00am = 8:00pm (Sydney time).
 - Answer: 8:00pm.

8. Work with 24-Hour Time in Timetable Questions

Strategy: Understand Timetable Formats (24-Hour Clock)

- How: In timetable questions, especially those involving 24-hour clocks, break down times carefully. If you see times like 14:30 or 21:45, convert them to 12-hour format to understand the duration between events.
- Example:
Question: "The train departs at 15:10 and arrives at 17:55. How long is the journey?"
 - Convert to 12-hour format:
 - 15:10 → 3:10pm
 - 17:55 → 5:55pm
 - The journey lasts 2 hours 45 minutes.
 - Answer: 2 hours 45 minutes.

Tables & Data Handling - Strategies

Questions involving tables and data handling test your child's ability to interpret data from different sources, such as bar charts, pie charts, Venn diagrams, and Carroll diagrams. These questions often ask for calculations like averages, finding missing values, or comparing fractions and percentages.

Here are strategies to solve these types of questions:

1. Understanding and Comparing Fractions

Strategy: Convert Fractions to the Same Denominator

- How: When comparing fractions with different denominators, you need to find a common denominator. The easiest way is to find the Lowest Common Denominator (LCD). This will allow you to compare the fractions more easily.
- Example:
Question: "If $\frac{1}{8}$, $\frac{1}{3}$, and $\frac{1}{4}$ represent fractions of a class, and we know the total number of pupils is 36, how do we find the number of pupils who like purple?"
 - The fractions are $\frac{1}{8}$, $\frac{1}{3}$, and $\frac{1}{4}$.
 - Find the Lowest Common Denominator (LCD):
The LCD of 8, 3, and 4 is 24.
 - $\frac{1}{8}$ becomes $\frac{3}{24}$
 - $\frac{1}{3}$ becomes $\frac{8}{24}$
 - $\frac{1}{4}$ becomes $\frac{6}{24}$
 - Add the fractions:
 $\frac{3}{24} + \frac{8}{24} + \frac{6}{24} = \frac{17}{24}$
 - The remaining fraction is:
 $\frac{24}{24} - \frac{17}{24} = \frac{7}{24}$.
 - Now, multiply by the total number of pupils:
 $\frac{7}{24} \times 36 = 7 \times 1.5 = 10.5$.
 - Answer: There are approximately 11 pupils who like purple. (Note: round to the nearest whole number if necessary).

2. Converting Fractions to Percentages

Strategy: Multiply the Fraction by 100

- How: To convert a fraction to a percentage, multiply the fraction by 100. This is especially helpful when interpreting data shown in pie charts or bar graphs.
- Example:
Question: "If $\frac{1}{4}$ of the class likes red, what percentage of the class likes red?"
 - To convert $\frac{1}{4}$ into a percentage, multiply by 100:
 $\frac{1}{4} \times 100 = 25\%$.
 - Answer: 25% of the class likes red.

3. Calculating Averages from Data

Strategy: Find the Mean by Adding All Values and Dividing by the Total Number

- How: When calculating averages (mean), add up all the data points and then divide by how many data points there are.
- Example:
Question: "The heights of five pupils are 140 cm, 145 cm, 150 cm, 155 cm, and 160 cm. What is the average height?"

- Add up all the heights:
 $140 + 145 + 150 + 155 + 160 = 750 \text{ cm.}$
- Divide by the number of pupils (5):
 $750 \div 5 = 150 \text{ cm.}$
- Answer: The average height is 150 cm.

4. Interpreting Bar Charts and Pie Charts

Strategy: Look at the Scale and Data Labels

- How: When reading bar charts or pie charts, check the scale (in the case of bar charts) or the percentages (in pie charts) to interpret the data. Ensure you understand what each bar or slice represents and use the labels or the scale to find the values you need.
- Example:
Question: "A pie chart shows the distribution of favourite colours in a class. If red represents 30%, green represents 25%, and blue represents 20%, what percentage of the class likes yellow?"
 - The total percentage is 100%.
 - Add the percentages for red, green, and blue:
 $30\% + 25\% + 20\% = 75\%.$
 - Subtract from 100% to find the percentage for yellow:
 $100\% - 75\% = 25\%.$
 - Answer: 25% of the class likes yellow.

5. Using Venn Diagrams

Strategy: Count Overlapping and Non-Overlapping Sections Carefully

- How: In Venn diagrams, ensure you count both the individual and overlapping sections correctly. The numbers in the overlapping sections represent items that belong to both categories.
- Example:
Question: "In a class of 30 pupils, 15 like soccer, 10 like basketball, and 5 like both. How many like only soccer?"
 - The total number of pupils who like soccer = 15.
 - 5 like both sports.
 - Therefore, the number of pupils who like only soccer = $15 - 5 = 10.$
 - Answer: 10 pupils like only soccer.

6. Carroll Diagrams

Strategy: Understand the Criteria for Each Section

- How: Carroll diagrams often have criteria to sort numbers into different categories (e.g., odd/even, greater than 10, divisible by 5, etc.). Make sure you understand the categories, and place the numbers accordingly.
- Example:
Question: "Sort the following numbers into a Carroll diagram with categories 'greater than 10' and 'divisible by 5': 5, 12, 15, 18, 20.
 - "Greater than 10": 12, 15, 18, 20
 - "Divisible by 5": 5, 15, 20
 - Numbers that are both greater than 10 and divisible by 5: 15, 20.
 - Answer: The numbers that are both greater than 10 and divisible by 5 are 15 and 20.

7. Handling Missing Data in Tables

Strategy: Use the Known Totals to Find Missing Values

- How: If there is missing data in a table, use the total number of items or the sum of the data to figure out the missing numbers.
- Example:
Question: "The table shows how many pupils prefer different subjects, but one value is missing. The total number of pupils is 40. The pupils who prefer Math is 12, Science is 14, and English is missing. What is the number of pupils who prefer English?"
 - Add up the known values:
 $12 \text{ (Math)} + 14 \text{ (Science)} = 26$.
 - Subtract from the total (40):
 $40 - 26 = 14$.
 - Answer: 14 pupils prefer English.
-